Beyond One Shot: Contracting for Sequential Trials

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Abstract

We study contract design in a setting where an agent can adaptively choose among several costly actions and resample the same action, while the principal observes only the final reported outcome. Given a contract, the agent's best response has a reservation-value structure. Unlike Weitzman's classic search problem, however, these reservation values are *stage-dependent* because the agent's willingness to continue depends on how many trials remain. Our main result is that a calibrated debt contract, which pays the agent only for outcomes above a face value set at the highest Weitzman index, nearly solves this dynamic incentive problem. Its payoff loss relative to the second best decays *exponentially* in the trial cap. In the infinite-trial benchmark, the same contract is exactly optimal: it implements the first-best value and leaves the agent with zero expected rent. We also show that more experimentation opportunities need not benefit either party. Increasing the trial cap can expand the agent's hidden continuation opportunities enough that the re-optimized second-best contract may simultaneously lower *both* the principal's and the agent's equilibrium payoffs. Finally, we extend the analysis to time-contingent payments, correlated sampling, and multiple agents, where our main insight about the asymptotic optimality of a calibrated debt contract still holds.

1 Introduction

Contracting under moral hazard, where an agent takes a costly, unobservable action and the principal observes only a stochastic performance signal, has been studied for nearly half a century (Holmström, 1979; Holmström and Milgrom, 1987). The theory is foundational because it provides a framework for designing incentives when effort is difficult to monitor, with applications ranging from executive compensation and corporate governance to sales-force compensation, procurement, outsourcing, insurance, risk sharing, and performance-based regulation.

Despite this breadth, much of this literature studies settings in which the agent has one single opportunity to act. Historically, this assumption was reasonable because many canonical applications are naturally modeled as one-shot (or effectively static) relationships. This premise, however,

is increasingly challenged by the rise of Large Language Models (LLMs) and AI systems, in which unobservable actions are repeated, low-cost, and rapidly adjustable. LLMs generate stochastic outputs, and even a simple instruction such as "Let's try again" can improve performance by eliciting additional rounds of reasoning (Liu et al., 2025). An agent can therefore privately rerun a prompt, revise instructions, invoke tools, or switch among models before delivering the final output. This shift is already reshaping pricing across industries, including professional services: clients are increasingly questioning billable hours as a measure of value when AI performs tasks such as data analysis and diagnosis, and are pushing firms such as McKinsey toward fees tied to realized outcomes.[1] These settings call for a moral-hazard model in which the agent takes *sequential* hidden actions, may repeat the same action because outputs are stochastic, observes intermediate outcomes privately, and ultimately reports one single final outcome to the principal.

We study this problem in a model where the principal commits ex ante to an outcome-contingent payment rule, subject to limited liability. The agent has a finite set of actions, each with a known sampling cost and a known action-specific outcome distribution. To capture practical constraints such as limited patience, deadlines, or resource budgets, we allow an exogenous cap on the number of samples the agent may take; this entails no loss of generality because the cap may be infinite. The agent chooses actions adaptively across trials: she may resample the same action many times or switch across actions as she observes outcomes. After stopping, she reports one realized outcome from her history, and the principal observes and verifies only that reported outcome. Thus, when only one trial is available, the model reduces to the canonical one-shot moral-hazard problem; with multiple trials, the same static-looking contract induces a dynamic search problem for the agent.

Our first result characterizes the agent's best response given an arbitrary contract. The optimal policy has an index-rule form. At any state, each action has a reservation value: the current best payment level at which the agent is indifferent between stopping and taking one more sample of that action. The agent stops when her current best payment exceeds all action-specific reservation values; otherwise, she continues with an action that maximizes the marginal gain from sampling. Unlike the stationary indices in Weitzman's classic search problem (Weitzman, 1979), these reservation values depend on the number of remaining trials. Intuitively, the agent is more willing to continue early in the process because she still has many future chances to improve on her current best payment; this raises the reservation value needed to make her stop. In the infinite-horizon benchmark, where the agent can keep sampling indefinitely, this dependence on the remaining trial budget disappears, and the reservation values reduce to stationary Weitzman cutoffs.

This stage dependence makes the principal's second-best problem difficult. A single outcome-contingent payment must control the agent's entire adaptive policy, not just one action choice. Changing the payment for an outcome can affect stopping and action selection at many remaining-trial levels simultaneously. The principal must therefore manage several incentive constraints at

[1] See Louise Lucas, "How AI is forcing McKinsey and its peers to rethink pricing," *Financial Times*, May 24, 2026.

once: productive search from low states, avoidance of excessive continuation after moderate successes, and the choice of which sampling distribution to use at each stage.

Our main result is that a simple debt contract nearly solves this dynamic incentive problem. The contract sets its face value at the largest Weitzman reservation value among the available actions. It then pays the agent nothing for outcomes below this face value and pays only the residual value above it. Under this contract, the agent repeatedly samples an action with the highest Weitzman index until either a draw reaches the face value or the trial cap is exhausted. We show that **this calibrated debt contract is asymptotically optimal with an exponential convergence rate**. For any finite trial cap, its payoff loss relative to the second-best contract is bounded by the probability that repeated sampling of a highest-index action fails to reach the face value before the cap is exhausted. This failure probability falls exponentially with the trial cap. In the infinite-horizon benchmark, the same contract is exactly optimal: it implements the first-best value and leaves the agent with zero expected rent. For any finite trial cap, it also leaves the agent with zero expected rent, but no longer necessarily implements the first best. Thus, the departure from the canonical one-shot moral-hazard benchmark is not rent extraction per se, but what repeated experimentation does to this zero-rent contract. In contrast, commonly used linear contracts are not asymptotically optimal and can perform arbitrarily poorly relative to the calibrated debt contract when the trial cap is large.

We also analyze optimal debt contracts for finite trial caps. When the number of trials is limited, the principal may improve within the debt family by choosing a lower face value, which makes it easier for the agent to reach the threshold before the cap is exhausted. For any fixed trial cap, we provide an exact polynomial-time algorithm for computing the optimal debt contract by exploiting the piecewise linearity of the principal's payoff in the debt face value. We then show that the optimal face value is never above the maximal Weitzman index, and that both the face-value gap and the payoff gain from re-optimizing within the debt family vanish exponentially fast as the trial cap grows. Thus, the simple calibrated debt contract rapidly becomes indistinguishable from the optimal finite-cap debt contract.

The trial cap itself has subtle payoff effects. More trials expand the agent's feasible set, so one might expect both parties to benefit. This is true under any fixed debt contract: increasing the trial cap weakly increases both the agent's and the principal's payoffs. Under re-optimized debt contracts, the principal's payoff remains weakly increasing, although the agent's payoff need not be monotone because the principal can respond to greater search flexibility by raising the face value. Outside the debt family, even the principal's payoff can fall. We exhibit a **lose-lose phenomenon** in which increasing the trial cap simultaneously lowers both the principal's and the agent's equilibrium payoffs under the second-best contract. The reason is that a larger trial cap gives the agent more hidden continuation opportunities, expanding the set of dynamic deviations that must be controlled using only the final reported outcome. The principal may

respond by re-optimizing the contract to discipline these deviations, but doing so can weaken productive experimentation enough to lower both the agent's and the principal's second-best payoff.

Finally, we study three extensions. First, we allow the principal to condition payments on both the reported outcome and the stopping time. This richer contract class restores monotonicity of the principal's second-best payoff in the trial cap, while the calibrated debt contract retains its asymptotic guarantee even relative to this larger class. Second, we allow correlated sampling in an infinite-horizon Markov environment, where the distribution of the next draw can depend on the current best outcome. A debt contract remains optimal under a natural monotonicity condition on the first-best value function. Third, we consider multiple agents and show that sequentially offering agent-specific calibrated debt contracts delivers an asymptotic guarantee. Together, these extensions show that the debt logic is not tied to the baseline independent-sampling, single-agent formulation.

2 Related Literature

Canonical moral-hazard models study settings in which a principal offers an outcome-contingent contract, an agent privately chooses a costly action or effort level, and the realized output provides only a noisy signal of that hidden action (Holmström, 1979; Grossman and Hart, 1992). These models are foundational because they formalize the central tradeoff between risk sharing and incentive provision, and because their tools—participation constraints, incentive-compatibility constraints, and first-order methods—remain central to modern contract design (Rogerson, 1985; Laffont and Martimort, 2009). Subsequent work extends this framework across a wide range of settings; see Georgiadis (2022) for a recent survey of moral-hazard models and methods, and Dütting et al. (2024) for the algorithmic contract-design literature.

One stream that is relevant to our work is dynamic moral hazard, which extends static principal–agent models to intertemporal environments. In these models, the agent repeatedly chooses costly hidden actions that affect the distribution of current or future output, while the contract uses the history of realized output to adjust current compensation and promised future utility (Spear and Srivastava, 1987; Sannikov, 2008). Dynamic moral hazard has been applied to operations settings in which the agent controls a single stochastic technology through unobserved actions: it is typically either a binary effort choice that shifts the rate of a Poisson arrival process for good or bad events, or an effort intensity that determines an arrival or breakthrough rate (Sun and Tian, 2018; Chen et al., 2020; Cao et al., 2024). Recent work extends this agenda to environments with learning about the agent's productivity (Wang et al., 2024) and adverse selection over the agent's type (Tian et al., 2025). In contrast, the dynamics in our model arise entirely on the agent's side: a fixed outcome-contingent payment must induce an experimentation policy that determines which action to sample, whether to resample or switch, and when to stop under a hard total trial cap.

The combination of an ex ante outcome-contingent contract and dynamic agent exploration connects our paper to delegated search, in which an agent privately searches over alternatives and the principal observes only the option ultimately proposed (Armstrong and Vickers, 2010; Lewis, 2012; Ulbricht, 2016). One branch of this literature guides the agent without direct monetary transfers. Acceptance and eligibility rules restrict which proposals can be implemented (Armstrong and Vickers, 2010; Kleinberg and Kleinberg, 2018), while information policies and acceptance standards can motivate continued search when transfers are unavailable (Xiao et al., 2026). Delegated-search models with monetary rewards also use payments to influence the agent's behavior, but they usually influence different choices from the ones we study. Lewis (2012) studies continuous-time delegated search in which the agent's hidden effort controls the arrival rate of discoveries, so rewards discipline search intensity and discovery timing. Erat and Krishnan (2012) study delegated design contests in which awards and problem specification affect participation and where agents search in a design space. Ulbricht (2016) considers an agent who privately knows the payoff distribution and privately chooses how long to sample from it. Zorc et al. (2025) study outsourcing in a finite-horizon sequential-search task where the agent decides whether to incur a search cost to draw a candidate from a single fixed distribution and whether to submit the candidate for costly evaluation by the principal. These papers show how rewards can shape search effort, participation, duration, discovery timing, or submission. In contrast, the agent in our model chooses among multiple actions, each with its own sampling distribution, and may resample actions. Thus, incentives in our model must govern not only whether and when the agent searches or submits, but also which distribution the agent samples at each stage of a finite trial process.

Given a contract, the agent's best-response problem in our setting is a Pandora-style sequential-search problem. In the classic Pandora's box model, an agent sequentially inspects costly boxes with recall, and the optimal policy is characterized by stationary reservation values (Weitzman, 1979). Subsequent work studies variants with more general payoff criteria (Olszewski and Weber, 2015), precedence or order constraints (Boodaghians et al., 2023), nonobligatory inspection (Doval, 2018), correlated alternatives (Brown and Smith, 2013), and Markovian items and capacity constraints (Brown and Smith, 2020). Other work uses Pandora-style search ideas in richer inspection and market-design environments. Aouad et al. (2026) study a Pandora's-box variant in which a searcher can first pay for partial inspection, observe a signal about a box's value, and then decide whether to pay for full inspection. Alaei et al. (2025) study an auction problem in which buyers may inspect an item to learn their private values. They show that this problem can be interpreted as a variant of Weitzman's Pandora's box problem with nonobligatory inspection. Papanastasiou and Sekar (2026) study product rankings for Pandora-style consumer search: rankings determine the order in which consumers inspect products and, through induced demand, whether sellers participate. The agent's search problem in our model adds two features that are central for incentives: the ability to resample the same action and a hard cap on the total number of trials. Together, these

features make the relevant reservation values stage dependent, because the value of sampling an action depends on the remaining trial budget and the agent's current search history.

Finally, our work is closest to recent papers on *contract design for sequential search*. Ezra et al. (2026) study a model in which the principal offers a payment rule over final outcomes and, given the contract, the agent's best response is a Pandora's box problem. In their model, there is no hard total trial cap, and each action can be taken at most once. They show that an optimal linear contract can be computed in polynomial time, while computing an optimal general contract is NP-hard when the number of outcomes is unrestricted. Hoefer et al. (2025) analyze a related model in which the final selected box and outcome are observed, so transfers can depend on the box-specific outcome rather than only on a common reported outcome. With this additional contractible information, they provide a polynomial-time algorithm for computing the optimal contract. Relative to both papers, our model allows resampling of the same action and adds a hard total trial cap, which changes the agent's induced search problem from a Weitzman-style index problem with fixed reservation values to one with stage-dependent reservation values.

Durandard et al. (2025) are especially relevant because they also obtain debt-like optimal contracts in a sequential-search environment. Their mechanism is robustness to the misspecification of the agent's feasible projects. In the spirit of Carroll (2015), the principal initially knows only *one* action and evaluates contracts by their worst-case performance over larger feasible action sets. They show that a debt contract with face value equal to the Weitzman index of the known action is robustly optimal. Our force is different. In their model, the robust payoff benchmark is pinned down by the singleton environment containing only the known action: the optimal debt contract ensures that additional unknown projects cannot reduce the principal's worst-case payoff. Our setting is different because the action set is known and contains multiple actions, so the central issue is not robustness to hidden alternatives but the dynamic incentive problem created by finite experimentation and resampling. A single outcome-contingent payment must guide not only whether the agent searches, but also which action she samples, whether she resamples or switches, and when she stops. This makes the relevant search incentives history- and stage-dependent. As the trial cap increases, these dynamic deviations become more difficult for the principal to control. A debt contract removes the main sources of dynamic over-search and premature stopping by paying the agent only for outcomes above a calibrated threshold. We show that a debt contract calibrated to the maximal Weitzman index resolves this dynamic incentive problem asymptotically: it induces repeated sampling of a maximal-index action and achieves a payoff loss that decays exponentially in the trial cap.

3 Model Setup

A *principal* contracts with an *agent* who can adaptively choose a sequence of costly actions, including repeated sampling of the same action, and then disclose one realized outcome to the principal. This setting captures operational environments in which an agent iteratively generates alternatives and ultimately selects one realized outcome for disclosure. Examples include repeated queries to a stochastic generative-AI pipeline, possibly using the same prompt–model pair; repeated simulation runs, possibly under a fixed design configuration; and repeated executions of the same production or service process when performance is stochastic.

There is a finite set of available actions, $\mathcal{A} := \{1, \dots, n\}$. Each time the agent takes action $i \in \mathcal{A}$, she incurs cost $c_i > 0$ and observes a stochastic outcome $Y \in \mathcal{Y} := \{1, \dots, m\}$ drawn from a known distribution $p_i = (p_{iy})_{y \in \mathcal{Y}}$, where $p_{iy} := \Pr(Y = y \mid \text{action } i)$ and $1^\top p_i = 1$. Resampling of the same action is allowed. We assume that each time action i is sampled, its outcome is drawn independently from the same distribution p_i. Conditional on the agent's chosen action sequence, outcomes are independent across trials. This assumption is relaxed in Section 6.2.

The agent faces a total trial budget $K \geq 1$: while resampling is allowed, the overall number of trials across all actions cannot exceed K. For example, when $K = 2$, the agent may sample action 1 twice, sample action 2 twice, or sample each action once. Economically, K captures a hard operational limit on experimentation, arising from factors such as the principal's limited patience, a deadline, or the agent's own resource constraints. When $K = 1$, so that the agent can take only one action, the model reduces to the canonical one-shot moral-hazard setting (Holmström, 1979).

At trial $k \in \{1, \dots, K\}$, the agent chooses an action $a_k \in \mathcal{A}$, incurs cost c_{a_k}, and observes an outcome $Y_k \sim p_{a_k}$. Let $h_k := (a_1, Y_1, \dots, a_k, Y_k)$ denote the history after k completed trials, with $h_0 := \emptyset$. A *policy* σ specifies, for each history h_k with $k \in \{0, 1, \dots, K-1\}$, whether the agent stops or continues and, if she continues, which action $a_{k+1} \in \mathcal{A}$ to take next. Let $\tau \in \{0, 1, \dots, K\}$ denote the endogenous stopping time induced by σ. Recall is allowed: if the agent stops at time $\tau \geq 1$, she may disclose any outcome in $\{Y_1, \dots, Y_\tau\}$.[2]

If outcome $y \in \mathcal{Y}$ is ultimately reported, the principal obtains value $v(y) \geq 0$. We assume that the model primitives $(\mathcal{A}, \mathcal{Y}, \{c_i, p_i\}_{i \in \mathcal{A}}, K, v)$ are common knowledge. The final reported outcome is verifiable ex post, but in the baseline model the principal does not observe the agent's action choices, interim outcomes, or stopping time. The principal therefore commits ex ante to an outcome-contingent payment *contract* $w : \mathcal{Y} \to \mathbb{R}_+$ satisfying limited liability, that is, $w(y) \geq 0$ for all y, as standard in the literature (Holmström, 1979; Carroll, 2015). In the baseline model, contracts can depend only on the final reported outcome and cannot condition on the agent's action history or stopping time. This is relaxed in Section 6.1.

[2] Our main result, Theorem 1, does *not* rely on the recall assumption.

Given (w, σ) and a realized history, the agent reports an outcome that maximizes her payment:

$$\hat{Y}(w, \sigma) \in \underset{t \in \{1,\dots,\tau\}}{\operatorname{argmax}} w(Y_t).$$

If $\tau = 0$, no action is taken, and both parties receive zero by convention. The agent's outside option is normalized to zero, so this zero-payoff outcome also serves as the participation benchmark. Reporting ties are broken in the principal's favor. When multiple agent policies maximize the agent's expected utility under a given contract, we select, among these best responses, a principal-favorable policy.

Both the principal and the agent are risk neutral. For any contract–policy pair (w, σ), the agent's expected utility is

$$u_A(w, \sigma) := \mathbb{E}\left[w\left(\hat{Y}(w, \sigma)\right) - \sum_{k=1}^{\tau} c_{a_k} \right],$$

and the principal's expected utility is

$$u_P(w, \sigma) := \mathbb{E}\Big[v\left(\hat{Y}(w, \sigma)\right) - w\left(\hat{Y}(w, \sigma)\right) \Big],$$

where the expectations are taken over the randomness in the outcomes $(Y_1, \dots, Y_\tau)$ generated along the action sequence $(a_1, \dots, a_\tau)$ induced by σ. For a given contract w, let $\sigma^*(w)$ denote a selected best response, that is, an agent-optimal policy maximizing $u_A(w, \sigma)$ over all policies with at most K trials, with the principal-favor selection rule above applied when the maximizer is not unique. The principal then chooses a contract w to maximize her equilibrium payoff $u_P(w, \sigma^*(w))$ induced by this selected best response.

The timeline of the game is as follows:

(i) The principal commits to a contract $w : \mathcal{Y} \to \mathbb{R}_+$.

(ii) The agent adaptively chooses actions for up to K trials, deciding after each observed outcome whether to stop or which action to take next (with resampling allowed).

(iii) Upon stopping, the agent, with recall, reports an outcome from her history that maximizes $w(\cdot)$; payments are then made and utilities are realized.

4 Agent's Best Response

In this section, we fix a contract $w : \mathcal{Y} \to \mathbb{R}_+$ and characterize a selected best-response policy $\sigma^*(w)$. Under a given contract, the agent chooses actions and a stopping time to maximize expected payment net of action costs. Because recall is allowed, whenever the agent stops, she reports an outcome in her history that maximizes $w(\cdot)$, with ties broken in the principal's favor. Accordingly, for the agent's stopping and action-selection problem, it is without loss of generality to summarize any history by two state variables: (i) the number of remaining trials and (ii) the current *best*

payment observed so far.

Formally, for $k \in \{0, 1, \ldots, K\}$ and $b \in \mathbb{R}_+$, let $V_k^w(b)$ denote the agent's optimal continuation value, that is, her maximal expected payoff from that point onward, when k trials *remain* and the best payment observed so far is b under contract w. These continuation values satisfy the following Bellman recursion:

$$V_0^w(b) = b, \tag{1}$$

$$V_k^w(b) = \max\left\{b,\ \max_{i\in\mathcal{A}}\left[-c_i + \mathbb{E}_{Y\sim p_i}\left[V_{k-1}^w\big(\max\{b, w(Y)\}\big)\right]\right]\right\}, \qquad k = 1, \ldots, K. \tag{2}$$

Equation (1) is the boundary condition: when no trials are left, the agent stops and receives the best payment secured so far. Equation (2) states that, at state (k, b), where $k \geq 1$ trials remain and b is the best payment observed so far, the agent either stops and receives b, or continues by choosing an action $i \in \mathcal{A}$, paying cost c_i, observing a fresh outcome $Y \sim p_i$, and obtaining the resulting expected continuation value, where the best running payment updates from b to $\max\{b, w(Y)\}$. In particular, the agent's expected utility at the start of the game, under contract w and best-response policy $\sigma^*(w)$, is $V_K^w(0)$.

At any state (k, b), the agent's *marginal gain* from choosing action $i \in \mathcal{A}$ rather than stopping is defined by

$$\Psi_k^w(b, i) := -c_i + \mathbb{E}_{Y\sim p_i}\left[V_{k-1}^w\big(\max\{b, w(Y)\}\big)\right] - b. \tag{3}$$

That is, $\Psi_k^w(b, i)$ is the net expected gain from continuing with action i rather than stopping immediately and securing b, given k trials remaining. This dynamic-programming formulation naturally leads to an action-specific cutoff. For each action $i \in \mathcal{A}$ and each $k \geq 1$, define the *reservation value* (or cutoff) for action i as the unique number z satisfying

$$-c_i + \mathbb{E}_{Y\sim p_i}\left[V_{k-1}^w\big(\max\{z, w(Y)\}\big)\right] = z, \qquad \text{equivalently,} \qquad \Psi_k^w(z, i) = 0. \tag{4}$$

We denote this cutoff by $z_k^w(i)$. It is the payment level at which the agent is exactly indifferent, with k trials remaining, between stopping and taking one more sample of action i. In particular, when b exceeds $z_k^w(i)$, action i is no longer worth sampling. Finally, define the maximal reservation value when k trials remain by

$$\bar{z}_k^w := \max_{i\in\mathcal{A}} z_k^w(i). \tag{5}$$

Thus, $\bar{z}_k^w$ is the highest reservation value across actions and determines the agent's stopping decision.

Lemma 1. *Fix any contract w. Then:*

(i) For each $k \in \{0, 1, \ldots, K\}$, the function $b \mapsto V_k^w(b)$ is nondecreasing, 1-Lipschitz, and convex on $\mathbb{R}$.

(ii) For every action $i \in \mathcal{A}$ and each $k \in \{1, \ldots, K\}$, the function $b \mapsto \Psi_k^w(b, i)$ is continuous,

nonincreasing, and convex on $\mathbb{R}$.

(iii) Equation (4) *admits a unique solution* $z_k^w(i) \in (-\infty, \bar{w})$*, where* $\bar{w} := \max\{w(y) : \exists i \in \mathcal{A} \text{ with } p_{iy} > 0\}$.

The monotonicity result in Lemma 1(i) reflects that the current best payment b is a guaranteed fallback: the agent can always stop and secure b. Consequently, the continuation value $V_k^w(b)$ is nondecreasing in b. The 1-Lipschitz property reflects that increasing the current best payment by δ raises the agent's fallback payoff by at most δ along any future sample path, so the continuation value can also increase by at most δ. Convexity comes from the stopping option embedded in the dynamic program: a higher b not only raises the payoff from stopping, but can also make stopping optimal in states where continuation was previously preferred. This policy-switching effect creates kinks and gives the value function a convex shape.

Lemma 1(ii) applies these properties to the marginal gain from taking one more sample of action i. The expected continuation payoff is nondecreasing and 1-Lipschitz in b; after subtracting the stopping payoff b, the marginal gain $\Psi_k^w(b, i)$ is therefore nonincreasing. Intuitively, as the current best payment b increases, there is less room for improvement, so the incremental value of additional sampling weakens. It is also continuous and convex. Convexity reflects that the marginal gain from sampling decreases less sharply as the current best payment rises. When b is low, increasing b mainly improves the stopping payoff, while the payoff after sampling is often determined by the new draw. When b is high, the same increase in b also carries into the post-sample state $\max\{b, w(Y)\}$, so it partly raises the continuation payoff as well. Thus sampling becomes less attractive as b rises, but at a decreasing rate.

The boundary behavior of $\Psi_k^w(\cdot, i)$ pins down the cutoff. For all $b \geq \bar{w}$, where $\bar{w}$ is the largest payment that can ever be realized under the contract, no future draw can improve the current best payment; one more trial only incurs the cost c_i, so $\Psi_k^w(b, i) = -c_i < 0$. Conversely, as $b \to -\infty$, the stopping payoff becomes arbitrarily unattractive while continuation remains bounded below, so $\Psi_k^w(b, i) \to +\infty$. Continuity therefore gives existence of a cutoff, and convexity together with strict negativity at $\bar{w}$ rules out more than one zero. Hence each action has a unique reservation value $z_k^w(i) < \bar{w}$. This reservation value may be negative: if sampling is already unprofitable at $b = 0$, then $z_k^w(i) < 0$, so action i is never worth sampling from any feasible state with $b \geq 0$.

Combining the Bellman recursion (2) with Lemma 1 yields the following characterization of the selected best-response policy.

Proposition 1 (Best-response policy)**.** *Fix any contract* w*. Consider any state* (k, b)*, where* $k \in \{1, \ldots, K\}$ *is the number of remaining trials and* $b \in \mathbb{R}_+$ *is the current best payment. Then:*

*(i) (*Stopping rule*) If* $b > \bar{z}_k^w$*, where* $\bar{z}_k^w$ *is defined in* (5)*, then stopping is optimal.*

*(ii) (*Continuation and action choice*) If* $b < \bar{z}_k^w$*, then it is optimal to continue and choose an*

action $i^*(k,b) \in \operatorname{argmax}_{i\in\mathcal{A}} \Psi_k^w(b,i)$*, with ties broken in the principal's favor.*

(iii) (Boundary tie) *If* $b = \bar{z}_k^w$*, the agent is indifferent between stopping and continuing with an action in* $\operatorname{argmax}_{i\in\mathcal{A}} \Psi_k^w(b,i)$*; ties are broken in the principal's favor.*

Proposition 1 gives an index-rule characterization of an agent's best-response policy. For each action i and remaining-trial level k, the reservation value $z_k^w(i)$ is the cutoff best-payment level at which the agent is indifferent between stopping and taking one more sample of action i. Thus action i is strictly worth sampling when the current best payment b is below $z_k^w(i)$; at equality the agent is indifferent. The stopping decision is governed by the largest reservation value $\bar{z}_k^w = \max_{i\in\mathcal{A}} z_k^w(i)$. If $b > \bar{z}_k^w$, no action yields a positive marginal gain, so stopping is optimal. If $b < \bar{z}_k^w$, some action yields a strictly positive marginal gain, so the agent continues and chooses an action maximizing $\Psi_k^w(b,i)$, with action ties broken in the principal's favor. If $b = \bar{z}_k^w$, the agent is indifferent between stopping and continuing with a maximal action, and the principal-favor tie-breaking rule selects among these agent-optimal choices. The case $k = 0$ is covered by the boundary condition $V_0^w(b) = b$: with no trials remaining, the agent must stop and secure the current best payment. Crucially, the indices are stage dependent: $z_k^w(i)$ depends on the number of remaining trials k. This dependence is the main departure from the stationary index rule.

Remark 1 (Relation to Weitzman's index rule)**.** *Weitzman (1979) studies a sequential search problem in which an agent sequentially inspects costly boxes, observes their realized values, and stops to keep the best realization obtained so far. Although the number of possible inspections is finite, because each box can be opened at most once, the optimal policy has a stationary index structure in the following sense: each unopened box is assigned a reservation value determined only by its own cost and value distribution. This reservation value is fixed throughout the search; it is not recomputed as a function of the stage, the current best realization, or the number of other boxes that remain unopened.*

Our model differs because the agent can resample actions but faces a total trial budget K*. The same action can therefore be sampled at different stages, and the value of sampling it depends on how many future trials remain after the current sample. Thus, for a fixed action* i*, the cutoff* $z_k^w(i)$ *is inherently* stage dependent*; indeed, it is weakly increasing in the number of remaining trials* k*, as shown in Proposition 2(i). When* $K = \infty$*, the dependence on remaining trials disappears, and the agent's best response reduces to a Weitzman-type stationary index rule; see Proposition 2(ii).*

Throughout, for any $x \in \mathbb{R}$, write $x^+ := \max\{x, 0\}$. With this notation in place, we can state the infinite-horizon counterpart of the finite-K cutoff characterization.

Proposition 2. *Fix any contract* w*.*

(i) For every action $i \in \mathcal{A}$*, the reservation value* $z_k^w(i)$ *is weakly increasing in the number of remaining trials:* $z_{k+1}^w(i) \geq z_k^w(i)$ *for* $k = 1, \ldots, K-1$*.*

(ii) Suppose $K = \infty$. For each action $i \in \mathcal{A}$, let $z^w_\infty(i)$ be the unique solution to

$$\mathbb{E}_{Y\sim p_i}[(w(Y) - z^w_\infty(i))^+] = c_i,$$

and define $\bar{z}^w_\infty := \max_{i\in\mathcal{A}} z^w_\infty(i)$. Then:

(a) Fix $i^*(w) \in \operatorname{argmax}_{i\in\mathcal{A}} z^w_\infty(i)$, with action ties broken in the principal's favor. An agent's best response is as follows:

(1) If $\bar{z}^w_\infty < 0$, stopping immediately is optimal.

(2) If $\bar{z}^w_\infty = 0$, the initial state $b = 0$ is a boundary state: the agent is indifferent between stopping and continuing with $i^*(w)$, and principal-favor tie-breaking selects between stopping and continuing with $i^*(w)$.

(3) If $\bar{z}^w_\infty > 0$, starting from $b = 0$, the agent samples $i^*(w)$ whenever the current best payment is strictly below $\bar{z}^w_\infty$. If the current best payment becomes strictly above $\bar{z}^w_\infty$, she stops; at equality, principal-favor tie-breaking selects between stopping and continuing with $i^*(w)$.

(b) The agent's expected utility is $u_A(w, \sigma^*(w)) = \max\{0, \bar{z}^w_\infty\}$.

Proposition 2(i) shows that the reservation values are weakly increasing in the number of remaining trials. Intuitively, sampling is more valuable when more trials remain because, after the current draw, the agent has more opportunities to improve on whatever best payment she has secured. Proposition 2(ii) identifies the stationary structure that emerges in the infinite-horizon benchmark. When $K = \infty$, the continuation problem no longer depends on a remaining trial budget, so each action has a time-invariant reservation value $z^w_\infty(i)$ determined by its cost and outcome distribution under the payment rule w. Because resampling is allowed, the agent can repeatedly draw from the same action. Hence, except at boundary states, she repeatedly samples a highest-reservation action below the cutoff $\bar{z}^w_\infty$ and stops above it; at equality, the principal-favor tie-breaking rule selects among the agent-optimal choices. This contrasts with the finite-K case in Proposition 1, where the cutoffs $z^w_k(i)$ depend on the remaining number of trials and the optimal action can vary across stages. Finally, the utility formula in Proposition 2(ii)(b) follows from the stationary value function. When $K = \infty$, the Bellman equation admits the closed form $V^w_\infty(b) = \max\{b, \bar{z}^w_\infty\}$; see the proof for details. Thus $\bar{z}^w_\infty$ is the agent's continuation value when the current best payment is below the stopping cutoff: it is the value of having unlimited opportunities to keep searching under the optimal stationary policy.

5 Debt Contracts

Characterizing the optimal contract is challenging. The principal chooses a single outcome-contingent payment rule, but for $K > 1$ that rule induces a full dynamic search policy: the agent decides when

to stop, which action to sample, whether to resample, and when to switch actions. By Proposition 1, these decisions are governed by stage-dependent reservation cutoffs. As a result, changing the payment rule can alter the agent's behavior at many remaining-trial levels simultaneously, so the principal's problem is not simply to induce one action but to control an entire adaptive stopping and sampling process.

We therefore focus on a specific contract class, *debt contracts*, and show that a carefully calibrated member of this class is asymptotically optimal. We use the standard limited-liability debt-contract form (Innes, 1990; Durandard et al., 2025). A debt contract with face value $x \geq 0$ pays

$$w^{\mathsf{debt}}(y; x) := \big(v(y) - x\big)^+, \qquad y \in \mathcal{Y}. \tag{6}$$

Under $w^{\mathsf{debt}}(\cdot; x)$, the principal keeps $\min\{v(y), x\}$ and the agent receives only the residual value above x. This terminology follows the standard corporate-finance analogy: the principal is a senior "debt holder" with promised face value x, while the agent is a residual "equity holder" who is paid only when the realized outcome $v(y)$ exceeds x. Thus low outcomes generate no payment to the agent, while high outcomes give the agent the residual surplus $v(y) - x$. We refer to $w^{\mathsf{debt}}(\cdot; x)$ as the x-debt contract throughout the rest of the analysis. For a trial cap K, let $\sigma^{*,K}(x)$ denote the selected agent's best response under this contract, and define the induced agent and principal payoffs by

$$U_A^{\mathsf{debt}}(K; x) := u_A\big(w^{\mathsf{debt}}(\cdot; x), \sigma^{*,K}(x)\big), \qquad U_P^{\mathsf{debt}}(K; x) := u_P\big(w^{\mathsf{debt}}(\cdot; x), \sigma^{*,K}(x)\big).$$

Let $\mathsf{SB}_P(K) := \max_{w:\mathcal{Y}\to\mathbb{R}_+} u_P(w, \sigma^*(w))$ denote the principal's second-best payoff under the optimal outcome-contingent contract, which we refer to as a *second-best contract*. Let $\mathsf{FB}_P(K)$ denote the corresponding first-best payoff: the payoff the principal would obtain if she directly controlled the adaptive sampling policy for up to K trials, observed all realized outcomes, kept the best realized outcome, and bore the sampling costs herself.

5.1 Asymptotic Optimality of the $\bar{\zeta}$-Debt Contract

We next focus on a particular debt contract, the $\bar{\zeta}$-debt contract, and show that it is asymptotically optimal. For each action $i \in \mathcal{A}$, let ζ_i denote the Weitzman reservation value, defined as the unique solution to

$$\mathbb{E}_{Y\sim p_i}\big[(v(Y) - \zeta_i)^+\big] = c_i. \tag{7}$$

This value depends only on the primitives of action i, i.e., its cost c_i and outcome distribution p_i, but not on the contract. Let

$$\bar{\zeta} := \max_{i\in\mathcal{A}} \zeta_i, \qquad \bar{\imath} \in \operatorname*{argmax}_{i\in\mathcal{A}} \zeta_i. \tag{8}$$

Assume $\bar{\zeta} > 0$; otherwise, the principal's second-best contract is to offer no contract, as shown in Equation (B-7) in the proof.

For convenience, under debt contracts we track the highest realized value rather than the highest payment so far. Because the agent's payment is weakly increasing in realized value, these two state variables are equivalent. Let $\bar{v}$ denote the highest realized value, with $\bar{v} = 0$ before any trial. Under $w^{\mathsf{debt}}(\cdot; \bar{\zeta})$, the current best payment is $(\bar{v} - \bar{\zeta})^+$.

Lemma 2 (Agent's best response under $w^{\mathsf{debt}}(\cdot; \bar{\zeta})$)**.** *Under the $\bar{\zeta}$-contract $w^{\mathsf{debt}}(\cdot; \bar{\zeta})$, at any state $(k, \bar{v})$ with $k \in \{1, \dots, K\}$ trials remaining and the current highest realized value $\bar{v} \geq 0$, an agent's best response is:*

(i) If $\bar{v} \geq \bar{\zeta}$, the agent stops.

(ii) If $\bar{v} < \bar{\zeta}$, continuing with action $\bar{i}$ is optimal. After observing $Y \sim p_{\bar{i}}$:

(a) if $v(Y) \geq \bar{\zeta}$, then the agent stops and reports this outcome;

(b) if $v(Y) < \bar{\zeta}$, then the highest realized value remains below $\bar{\zeta}$, and the agent repeats action $\bar{i}$ whenever another trial remains.

In particular, starting from the initial state, the agent repeatedly samples $\bar{i}$ until either some draw yields $v(Y) \geq \bar{\zeta}$ or the cap K is exhausted.

Lemma 2 shows that the $\bar{\zeta}$-debt contract has a simple and robust behavioral implication: even with a finite trial budget, the agent never switches actions. Starting from the initial state, she repeatedly samples $\bar{i}$, an action attaining the maximal Weitzman index $\bar{\zeta}$, until either a draw reaches the face value, i.e., $v(Y) \geq \bar{\zeta}$, or the cap is exhausted. This no-switching property is striking. In the finite-K problem, the agent's reservation cutoffs generally depend on the number of remaining trials (cf. Proposition 1), so one would expect the optimal action to vary across stages. Indeed, even within the debt-contract family $w^{\mathsf{debt}}(\cdot; x)$, such switching typically arises for a generic face value $x < \bar{\zeta}$. The $\bar{\zeta}$-debt contract avoids this complication because its face value is calibrated to the maximal Weitzman index. For every action i,

$$\mathbb{E}_{Y \sim p_i}\left[(v(Y) - \bar{\zeta})^+\right] \leq c_i,$$

with equality only for actions in $\operatorname{argmax}_j \zeta_j$. Under $w^{\mathsf{debt}}(\cdot; \bar{\zeta})$, when the current best payment is zero, one more sample has zero net value for maximal-index actions and negative net value for all other actions. More generally, $V_k(b) = b$ for all k and $b \geq 0$, so the agent's continuation value is pinned down by the current best payment b. The induced behavior is easiest to describe in terms of the highest realized value $\bar{v}$. If $\bar{v} > \bar{\zeta}$, the current best payment is positive and stopping is optimal. If $\bar{v} = \bar{\zeta}$, stopping gives the principal payoff $\bar{\zeta}$ and is selected by principal-favor tie-breaking. If $\bar{v} < \bar{\zeta}$, the agent is indifferent between stopping and sampling a maximal-index action, and principal-favor tie-breaking selects sampling $\bar{i}$, a maximal-index action, whenever another trial remains. The same

reasoning also explains why other debt contracts need not have this no-switching property. If the face value is set below $\bar{\zeta}$, some action can generate strictly positive continuation surplus when the current best payment is low. With a finite trial budget, the value of that surplus depends on how many trials remain, so the action that maximizes continuation value can vary across stages, leading to switching.

Finally, define the per-trial probability that action $\bar{i}$ reaches the face value $\bar{\zeta}$ by

$$\bar{q} := \Pr_{Y \sim p_{\bar{i}}} \{v(Y) \geq \bar{\zeta}\}. \tag{9}$$

Theorem 1 (Asymptotic optimality of $w^{\mathsf{debt}}(\cdot; \bar{\zeta})$)**.** *For every trial cap $K \geq 1$,*

$$\frac{U_P^{\mathsf{debt}}(K; \bar{\zeta})}{\mathsf{SB}_P(K)} \;\geq\; 1 - (1 - \bar{q})^K. \tag{10}$$

Moreover:

(i) In the infinite-horizon benchmark $K = \infty$, the $\bar{\zeta}$-debt contract is first-best optimal:

$$\mathsf{FB}_P(\infty) = \mathsf{SB}_P(\infty) = U_P^{\mathsf{debt}}(\infty; \bar{\zeta}) = \bar{\zeta}.$$

(ii) Under the $\bar{\zeta}$-debt contract, the agent's expected payoff is zero for every trial cap, finite or infinite: $U_A^{\mathsf{debt}}(K; \bar{\zeta}) = 0$ for all integers $K \geq 1$.

Theorem 1 gives a finite-K performance guarantee for the $\bar{\zeta}$-debt contract. By Lemma 2, the contract induces repeated sampling of the maximal-index action $\bar{i}$. Thus, after K trials, the probability that at least one draw reaches the face value $\bar{\zeta}$ is $1-(1-\bar{q})^K$. On that event, an outcome with value at least $\bar{\zeta}$ is available at the reporting stage, so the principal obtains payoff $\bar{\zeta}$ under the debt contract. Since the second-best payoff is bounded above by the infinite-horizon first-best value $\bar{\zeta}$, the contract guarantees at least a $1 - (1 - \bar{q})^K$ fraction of the second-best payoff. Equivalently, the unguaranteed fraction is at most $(1 - \bar{q})^K$, which decays *exponentially* in K. The exponential rate makes the guarantee meaningful even for moderate trial caps K and $\bar{q}$. For example, when $K = 15$ and $\bar{q} = 0.1$, the guarantee is nearly 80%. Moreover, the actual performance of the $\bar{\zeta}$-debt contract is often better than the bound in (10) for finite K. The bound counts only the event that at least one draw reaches the face value $\bar{\zeta}$; even when no draw reaches the face value, the principal may still obtain a positive payoff from the best below-threshold realization.

The infinite-horizon result in Theorem 1(i) follows because the upper bound becomes tight. Since the infinite-horizon problem is stationary, each action has a fixed Weitzman reservation value. The value of the first-best self-search problem is the largest of these reservation values, $\bar{\zeta}$. This is because the first-best policy repeatedly samples a maximal-index action and stops once the current best realized value reaches the cutoff; the terminal realized value may exceed $\bar{\zeta}$, but the expected net value of this policy is exactly $\bar{\zeta}$. The $\bar{\zeta}$-debt contract reproduces the same cutoff logic by paying the agent only for outcomes above $\bar{\zeta}$, so the agent's privately optimal sampling and stopping behavior

implements the principal's stationary first-best rule. Hence $\mathsf{FB}_P(\infty) = \mathsf{SB}_P(\infty) = U_P^{\mathsf{debt}}(\infty; \bar{\zeta}) = \bar{\zeta}$. The same proof also yields the analogous guarantee relative to the finite-K first best:

$$\frac{U_P^{\mathsf{debt}}(K; \bar{\zeta})}{\mathsf{FB}_P(K)} \geq 1 - (1 - \bar{q})^K,$$

because $\mathsf{SB}_P(K) \leq \mathsf{FB}_P(K) \leq \mathsf{FB}_P(\infty) = \bar{\zeta}$. The inequality $\mathsf{FB}_P(K) \leq \mathsf{FB}_P(\infty)$ follows from the monotonicity of the first-best self-search problem: because the principal directly controls the sampling policy, any cap-K policy remains feasible when more trials are available by simply leaving the additional trials unused.

For Theorem 1(ii), the zero-rent result follows from the calibration of the face value. Under the $\bar{\zeta}$-debt contract, when the current best payment is zero, one more sample has weakly negative expected net value for every action and exactly zero net value for maximal-index actions. If $b > 0$, the current fallback further reduces the upside from sampling, so stopping is optimal. Hence the agent's continuation value is $V_k(b) = b$ at every state, and in particular $U_A^{\mathsf{debt}}(K; \bar{\zeta}) = V_K(0) = 0$. This zero-rent conclusion is not unique to large K. Even when $K = 1$, the $\bar{\zeta}$-debt contract makes the agent indifferent between taking a maximal-index action and taking the outside option, since $\mathbb{E}_{Y \sim p_{\bar{i}}}[(v(Y) - \bar{\zeta})^+] = c_{\bar{i}}$. Under principal-favor tie-breaking, the agent takes that action and earns zero expected payoff. Thus the departure from the canonical one-shot moral-hazard benchmark is not rent extraction per se, but what repeated experimentation does to this zero-rent contract. For finite K, zero rent does not imply first-best implementation. The $\bar{\zeta}$-debt contract induces a stationary policy: the agent repeatedly samples a maximal-index action $\bar{i}$. The finite-K first-best policy, however, may be stage dependent and may use different actions at different remaining-trial levels. With a limited number of trials, the principal may prefer actions with different risk–return tradeoffs than the infinite-horizon maximal-index action. As K grows, this gap becomes small because repeated samples of $\bar{i}$ reach the face value with probability converging to one at an exponential rate. In the infinite-horizon benchmark, the same cutoff logic exactly matches the stationary first-best search rule, so the $\bar{\zeta}$-debt contract implements the first-best value while still leaving the agent zero expected rent.

We now discuss the critical index value $\bar{\zeta}$ in more detail. Since $\bar{\zeta}$ is the maximum reservation value across actions, adding feasible actions while keeping the original actions fixed weakly increases $\bar{\zeta}$; that is, $\bar{\zeta}$ is weakly increasing in n. However, a larger $\bar{\zeta}$ does not by itself imply better finite-K performance. Under the $\bar{\zeta}$-debt contract, the principal secures the face value within K trials only if repeated samples of a maximal-index action reach the threshold, an event with probability $1 - (1 - \bar{q})^K$, where $\bar{q} = \Pr(v(Y_{\bar{i}}) \geq \bar{\zeta})$. Thus finite-$K$ guarantee depends both on the level of the maximal index and on the probability of crossing it. In contrast, the number of possible outcomes m has no monotone effect on $\bar{\zeta}$ by itself; what matters is how the outcome distribution changes the tail expectation $\mathbb{E}[(v(Y) - z)^+]$ that defines each action's reservation value.

The preceding result relies on the threshold structure of debt contracts. To see why this structure matters, we compare it with the canonical benchmark of *linear contracts* (Holmström and Milgrom, 1987; Carroll, 2015). A linear contract pays the agent a fixed share α of realized value:

$$w_\alpha^{\mathsf{lin}}(y) := \alpha\, v(y), \qquad \alpha \in [0,1].$$

For a trial cap K, let $U_P^{\mathsf{lin}}(K;\alpha) := u_P\left(w_\alpha^{\mathsf{lin}},\, \sigma^*(w_\alpha^{\mathsf{lin}})\right)$ denote the principal's expected utility under the linear contract w_α^{lin} when the agent best-responds. The best payoff attainable within the linear-contract class is $U_{P,*}^{\mathsf{lin}}(K) := \max_{\alpha\in[0,1]} U_P^{\mathsf{lin}}(K;\alpha)$.

Remark 2 (Linear contracts can be arbitrarily bad as $K \to \infty$)**.** *Fix any integer $K \geq 2$ and consider the following example: $\mathcal{A} = \mathcal{Y} = \{1,\dots,K\}$, deterministic outcomes $p_{ii} = 1$, values $v(y) = 2^y$, and costs $c_i = 2^i - i$. For this instance, $U_P^{\mathsf{debt}}(K;\bar\zeta) = K$. In contrast, the optimal linear contract and the associated principal payoff are*

$$\alpha^* = 1 - 2^{-(K-1)} \in (0,1), \qquad U_{P,*}^{\mathsf{lin}}(K) = 2.$$

Therefore,

$$\frac{U_{P,*}^{\mathsf{lin}}(K)}{U_P^{\mathsf{debt}}(K;\bar\zeta)} = \frac{2}{K} \to 0 \qquad \text{as } K \to \infty.$$

The proof is deferred to the appendix.

The example highlights a fundamental difference between linear and debt contracts. A linear contract pays the agent a constant fraction of the entire realized value. In this instance, values grow exponentially, $v(i) = 2^i$, while costs are $c_i = 2^i - i$, so the agent's private gain from moving to a higher action is small relative to the increase in total value. To make the agent choose the top action, a linear contract must therefore give her almost the entire value: comparing the agent's payoff from action K with that from action $K-1$ gives $\alpha 2^K - (2^K - K) \geq \alpha 2^{K-1} - (2^{K-1} - (K-1))$, or equivalently $1 - \alpha \leq 2^{-(K-1)}$. But then the principal keeps only the small residual share $1-\alpha$, so her payoff under the optimal linear contract is bounded by 2. Debt separates incentives from surplus extraction differently. The $\bar\zeta$-debt contract pays the agent only the residual value above the face value $\bar\zeta = K$. This residual claim is enough to make the top action privately attractive: action K gives the agent payment $2^K - K$, exactly offsetting its cost $2^K - K$. At the same time, the principal keeps the first K units of realized value, so her payoff is K. Thus linear sharing must give away almost all of the realized value to induce the top action, whereas debt can preserve the same incentive while letting the principal retain a growing face-value payoff.

5.2 Optimal Debt Contracts for Finite K

The preceding results show that the calibrated $\bar\zeta$-debt contract has strong large-K performance. For a fixed finite trial cap, however, the principal may be able to improve within the debt family by

choosing a face value different from $\bar{\zeta}$. We therefore next study the finite-K optimal debt contract: how to compute the best face value, and how far this optimized debt contract can be from the simple $\bar{\zeta}$-debt contract. Let $x^*_{\mathsf{debt}}(K) \in \operatorname{argmax}_{x\geq 0} U_P^{\mathsf{debt}}(K;x)$ be an optimal debt face value when the trial cap is K. In addition, denote

$$U_{P,*}^{\mathsf{debt}}(K) := \max_{x\geq 0} U_P^{\mathsf{debt}}(K;x)$$

as the principal's payoff under an optimal debt contract.

Proposition 3 (Computing the optimal finite-K debt contract)**.** *Fix $K \geq 1$. There is an algorithm that computes an optimal debt face value $x^*_{\mathsf{debt}}(K)$ in time $O(n^{3K+1}m^{K+1})$, which is polynomial in the number of actions n and the number of outcomes m.*

Proposition 3 shows that, for a fixed trial cap K, the optimal debt face value can be computed in polynomial time in the number of actions and outcomes; the exact algorithm is given in Appendix A. The debt structure avoids the usual state-space curse of dimensionality: once the face value x is fixed, the payoff-relevant state is only the current running maximum and the number of trials remaining. The main computational challenge is that, as x varies, the agent's optimal choice can switch between stopping and different actions. The algorithm must therefore track both the agent's payoff and the principal's payoff as functions of x throughout the dynamic program. For each value of x, it compares the payoff pairs generated by stopping and by each possible action: it first keeps the alternatives that maximize the agent's payoff, and then, among agent-payoff ties, keeps the alternative with the highest principal payoff. Because these candidate payoff pairs are *piecewise affine* in x, the selected payoff pair can change only at finitely many crossing points. Hence $U_P^{\mathsf{debt}}(K;x)$ is affine between finitely many breakpoints, and an optimal face value can be found by checking the final breakpoint list.

Remark 3. *The complexity is polynomial in (n,m) for any* fixed *K, but can grow exponentially in K because each dynamic-programming layer refines the breakpoint lists inherited from the previous layer and adds new crossing points where the agent's preferred continuation choice changes. This limitation is less restrictive for the regimes of interest here. For small K, exact optimization is computationally easy. For large K, Theorem 1 shows that the simple $\bar{\zeta}$-debt contract is already near-optimal, with an exponentially small performance loss, so exact finite-K optimization within the debt family is less important.*

Our next result shows that the optimal finite-K debt face value converges exponentially fast to $\bar{\zeta}$. Moreover, the payoff loss from using the simple $\bar{\zeta}$-debt contract rather than re-optimizing within the debt family vanishes at an exponential rate.

Proposition 4. *For every $K \geq 1$ and every optimal debt face value $x^*_{\mathsf{debt}}(K)$, we have*

$$0 \;\leq\; \bar{\zeta} - x^*_{\mathsf{debt}}(K) \leq\; \bar{\zeta}(1-\bar{q})^K,$$

and

$$0 \;\le\; U_{P,*}^{\mathsf{debt}}(K) - U_P^{\mathsf{debt}}(K;\bar{\zeta}) \;\le\; \bar{\zeta}(1-\bar{q})^K.$$

The proposition first shows that no optimal debt face value lies above $\bar{\zeta}$. Equivalently, the $\bar{\zeta}$-debt contract is weakly more *aggressive* than any optimal finite-K debt contract: it sets a higher face value, so the principal keeps a larger capped claim $\min\{v(y), x\}$ while the agent is paid only the residual upside $(v(y)-x)^+$. Raising x tightens incentives by shrinking the set of outcomes that generate a positive payment. Once $x > \bar{\zeta}$, this tightening goes too far: every action has strictly negative expected net value for the agent, so the agent stops immediately and the principal obtains zero. Thus $\bar{\zeta}$ is the largest face value that can sustain nontrivial search, making the $\bar{\zeta}$-debt contract the most aggressive debt contract that remains incentive compatible in this sense.

Second, although a lower face value can be useful for finite K, the value of such fine-tuning is small. Lowering the face value can make threshold crossing easier within a limited trial budget, but it also lowers the principal's payoff when the threshold is reached. The proof formalizes this tradeoff with two simple bounds. Under any x-debt contract, the principal's realized payoff is at most x. Meanwhile, the $\bar{\zeta}$-debt contract guarantees the principal an expected payoff of at least $\bar{\zeta}(1-(1-\bar{q})^K)$: with probability $1-(1-\bar{q})^K$, action $\bar{\imath}$ reaches the face value within K trials, and on that event the principal's realized payoff is $\bar{\zeta}$. Therefore any optimal face value must be at least this guaranteed payoff level and at most $\bar{\zeta}$, yielding the face-value bound.

The payoff-gap bound follows from the same comparison. The optimal debt payoff is at most $\bar{\zeta}$, while the $\bar{\zeta}$-debt contract guarantees $\bar{\zeta}(1-(1-\bar{q})^K)$. Hence the most the optimal finite-K debt contract can improve on the simple $\bar{\zeta}$-debt contract is

$$U_{P,*}^{\mathsf{debt}}(K) - U_P^{\mathsf{debt}}(K;\bar{\zeta}) \;\le\; \bar{\zeta}\cdot(1-\bar{q})^K.$$

Both gaps are therefore controlled by the probability $(1-\bar{q})^K$ that the maximal-index action fails to reach the face value within the trial cap. As K grows, the gain from solving the exact finite-K debt-optimization problem rather than using the simple $\bar{\zeta}$-debt contract becomes exponentially small.

5.3 How Trial Caps Shape Payoffs

We next examine how payoffs change as the trial cap expands. A larger cap gives the agent more flexibility: she can keep searching after an unsuccessful draw, potentially generating better outcomes. One might therefore expect additional trials to benefit both parties. This intuition is only partly correct as shown in Proposition 5.

Proposition 5 (Payoff monotonicity under debt contracts)**.** *We have*

(i) Fix any debt contract $w^{\mathsf{debt}}(\cdot;x)$ with face value $x \ge 0$. As the trial cap $K \ge 1$ increases, both

parties' payoffs under the agent's best-response policy weakly increase:

$$U_A^{\mathsf{debt}}(K+1;x) \geq U_A^{\mathsf{debt}}(K;x), \qquad U_P^{\mathsf{debt}}(K+1;x) \geq U_P^{\mathsf{debt}}(K;x).$$

(ii) Under optimal debt contracts with re-optimized face values $x^*_{\mathsf{debt}}(K)$, the principal's payoff is still weakly increasing in K:

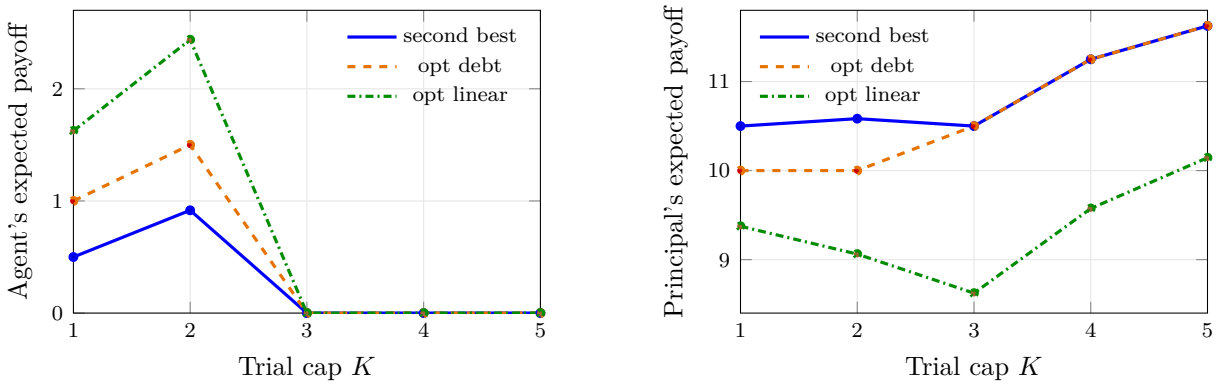
$$U_{P,*}^{\mathsf{debt}}(K+1) \geq U_{P,*}^{\mathsf{debt}}(K), \qquad K \geq 1.$$

However, the agent's payoff $U_A^{\mathsf{debt}}(K; x^*_{\mathsf{debt}}(K))$ may not *be monotone in K.*

Proposition 5 relies on a special feature of debt contracts that is absent from general contracts. Fix a face value x, and let $\bar{v}$ denote the current running maximum of realized values. Under the x-debt contract, the principal's realized payoff is $\min\{\bar{v}, x\}$, while the agent's realized payoff is $(\bar{v} - x)^+$ net of sampling costs. Thus the face value separates the payoff consequences of search into two regions. Below the face value, all outcomes with value less than x give the agent the same payment, zero. A moderately good draw below x therefore does not give the agent a new incentive to stop early; the agent is still searching for value above the threshold. From the principal's perspective, additional trials in this region can only raise the running maximum and hence weakly increase the capped payoff $\min\{\bar{v}, x\}$. Once the running maximum reaches the face value, the principal's payoff is already capped at x. Further experimentation may increase the agent's residual payoff above x, but it cannot reduce the principal's realized payoff. These two regions explain why, under a fixed debt contract, expanding the trial cap weakly benefits both the agent and the principal. The proof formalizes this intuition through continuation values. Let $A_k(\bar{v})$ and $P_k(\bar{v})$ denote the agent's and principal's continuation payoffs under principal-favor tie-breaking when k trials remain and the current running maximum value is $\bar{v}$. The agent's monotonicity is immediate from feasibility: with one more trial available, she can always follow her old policy and stop early. The principal's monotonicity is more specific to debt contracts. The proof shows that $P_k(\bar{v})$ is nondecreasing in $\bar{v}$ and that, for *every* state $(k, \bar{v})$, $P_{k+1}(\bar{v}) \geq P_k(\bar{v})$. Since the game starts from running maximum $\bar{v} = 0$ and $U_P^{\mathsf{debt}}(K;x) = P_K(0)$, applying this statewise inequality at $\bar{v} = 0$ yields $U_P^{\mathsf{debt}}(K+1;x) \geq U_P^{\mathsf{debt}}(K;x)$ for every fixed face value x.

The second part of Proposition 5 distinguishes fixed face values from re-optimized face values. Since the principal's payoff is weakly increasing in K for every fixed x, she can always reuse the old optimal face value when the cap expands; hence $U_{P,*}^{\mathsf{debt}}(K)$ is weakly increasing. The agent's payoff under the re-optimized face value $x^*_{\mathsf{debt}}(K)$ need not be monotone. As K grows, the principal may raise the face value in response to the agent's greater search flexibility, tightening the contract enough that the agent's equilibrium payoff falls. In the limit, Theorem 1 implies $x^*_{\mathsf{debt}}(K) \to \bar{\zeta}$ and the agent's expected payoff converges to zero.

The proposition establishes monotonicity for debt contracts, but it also raises a natural question:



(a) Agent's payoff

(b) Principal's payoff

Figure 1: Impact of the trial cap K on equilibrium payoffs

Note. Consider an instance with one agent, three actions, and three possible outcomes y_1, y_2, y_3, with principal values $v(y_1) = 0$, $v(y_2) = 12$, and $v(y_3) = 18$. The action distributions over the three outcomes, together with their costs, are $p_1 = (1/2, 1/3, 1/6)$ with $c_1 = 1$, $p_2 = (0, 1/2, 1/2)$ with $c_2 = 4$, and $p_3 = (3/5, 2/5, 0)$ with $c_3 = 2/5$.

whether similar monotonicity patterns arise under alternative contract classes. Figure 1 illustrates how the trial cap affects equilibrium payoffs across three contract classes: the unrestricted optimal contract (second best), the optimal debt contract, and the optimal linear contract. For each contract class, the contract parameters are re-optimized separately for each trial cap K. Panel (a) reports the agent's payoff, and Panel (b) reports the principal's payoff.

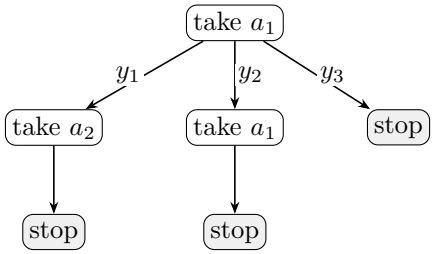
We first consider the agent's expected payoff, shown in Figure 1(a). For debt contracts, the figure is consistent with Proposition 5: once the face value is re-optimized for each K, the agent's payoff need not be monotone. The same pattern can arise under the second-best and optimal linear contracts. The reason is similar: a larger trial cap improves the agent's ability to search for good outcomes, but it also gives the principal room to tighten the contract and extract more of the resulting surplus. Thus, more flexibility for the agent need not translate into a higher equilibrium payoff.

Turning to the principal's expected payoff, Figure 1 is again consistent with Proposition 5: when the face value is re-optimized, the principal's payoff under debt contracts (weakly) increases with K. This monotonicity is special to debt contracts and does not generally extend to linear or second-best contracts. Under a debt contract, additional trials can only help the principal before the running maximum reaches the face value: the agent receives zero for all below-threshold outcomes, so moderate successes do not create a private incentive to stop early. Once the face value is reached, the principal's payoff is already capped and protected. This logic does not extend to linear or general contracts. An additional available trial can change the agent's action choice and continuation decision in ways that are privately optimal for the agent but reduce the principal's equilibrium payoff.

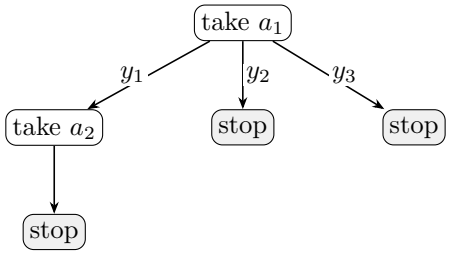
We next zoom in on the transition from $K = 2$ to $K = 3$ in Figure 1. We focus on the second-best contract to isolate the mechanism. This comparison makes concrete why increasing the trial cap can make both parties, and in particular the principal, worse off outside the debt-contract class. Table 1 reports one optimal contract in each contract class at $K = 2$ and $K = 3$, together with the corresponding equilibrium payoff pair (U_A, U_P) and its two-decimal approximation. For the second-best contract, w is the payment vector over outcomes with values $(0, 12, 18)$. For debt contracts, x is the face value. The primitives of the model imply Weitzman reservation values $\zeta_1 = 12$, $\zeta_2 = 11$, and $\zeta_3 = 11$, so $\bar{\zeta} = 12$. Thus, at $K = 3$, the optimal debt contract in the table already coincides with the $\bar{\zeta}$-debt contract. Figure 2 shows the agent's best-response decision trees, making explicit how the optimal contract changes the agent's continuation and stopping behavior as the trial cap increases.

Table 1: Optimal contracts and equilibrium payoffs at $K = 2$ and $K = 3$

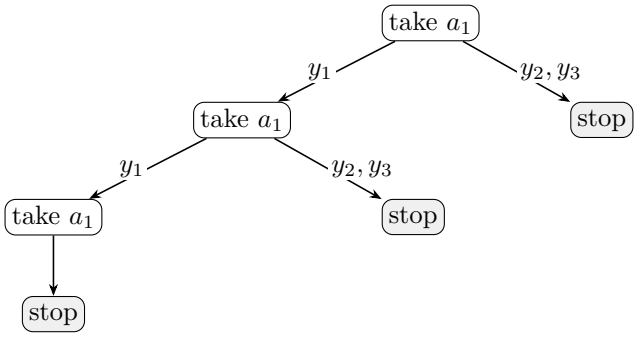
	$K = 2$	$K = 3$
Second best	$w = (0, 0, 9)$ $(U_A, U_P) = \left(\frac{11}{12}, \frac{127}{12}\right) \approx (0.92, 10.58)$	$w = (0, 0, 6)$ $(U_A, U_P) = \left(0, \frac{21}{2}\right) \approx (0, 10.50)$
Optimal debt	$x = 10$, $w = (0, 2, 8)$ $(U_A, U_P) = \left(\frac{3}{2}, 10\right) \approx (1.50, 10.00)$	$x = 12$, $w = (0, 0, 6)$ $(U_A, U_P) = \left(0, \frac{21}{2}\right) \approx (0, 10.50)$



(a) $K = 2$, second best



(b) $K = 2$, optimal debt



(c) $K = 3$, second best and optimal debt

Figure 2: Agent best-response decision trees for the contracts in Table 1.

Table 1 and Figure 2 highlight how a larger trial cap can make the agent's dynamic deviations

more costly for the principal to control. This is analogous to adding more actions in a static moral-hazard problem: the agent has more ways to respond to a given contract, so the principal must satisfy more incentive constraints. In this example, the agent can reach the moderate outcome y_2 while still having trials left under both $K = 2$ and $K = 3$. The key difference is that, when the cap increases, such histories become more important for incentive design: the agent has more opportunities to reach y_2 while still having continuation opportunities, making the temptation to gamble for y_3 harder to control. Because the contract cannot condition on the agent's history or remaining trial level, the principal cannot reward continuation after y_1 while discouraging gambling after y_2. Anticipating this larger set of dynamic deviations, she lowers the reward for the best outcome y_3. This curbs inefficient continuation after a moderate fallback, but it also weakens incentives to search from low states, where additional experimentation is valuable. This creates a *lose-lose* effect for both the agent and the principal.

Remark 4 (Lose-lose phenomenon)**.** *When the cap increases, the agent gains more flexibility. The principal may respond by re-optimizing the contract to control the expanded set of dynamic deviations, but this can weaken productive experimentation enough to lower* both *the agent's and the principal's second-best payoff.*

Debt contracts remove the main source of dynamic over-search and premature stopping. Before the threshold is reached, all below-threshold outcomes give the agent the same payment, zero; after the threshold is reached, the principal's payoff is already capped at the face value. Thus, the debt form limits the harmful dynamic deviations. The $\bar{\zeta}$-debt contract pushes this logic to the maximal sustainable threshold. Its face value is set at the maximal Weitzman index, so one more sample has weakly nonpositive net value for every action and exactly zero net value for maximal-Weitzman-index actions when the current payment is zero. As a result, the agent repeatedly samples a maximal-index action until a threshold-crossing outcome is found or the cap is exhausted.

6 Extensions

This section studies three extensions of the baseline model. Section 6.1 allows payments to depend on both the reported outcome and the stopping time, and shows that the main debt-contract guarantee is unchanged. Section 6.2 relaxes independent sampling by allowing the next draw to depend on the best outcome obtained so far. Section 6.3 extends the analysis to sequential contracting with multiple agents, where the principal may solicit several reports but ultimately adopts only the best one.

6.1 Time-Contingent Contracts

In the baseline model, the principal commits ex ante to an outcome-contingent payment rule $w : \mathcal{Y} \to \mathbb{R}_+$. This is appropriate when only the reported outcome is verifiable. In practice, however, the principal may also observe other dimensions of the search process, especially how long the agent worked on the task. We therefore consider a richer contract space in which payments can depend not only on the reported outcome but also on the stopping time $\tau \in \{0, 1, \dots, K\}$. Formally, a *time-contingent contract* is a collection

$$W := (w_t)_{t=1}^{K} \qquad \text{where} \qquad w_t : \mathcal{Y} \to \mathbb{R}_+,$$

satisfying limited liability $w_t(y) \geq 0$ for every stopping time t and outcome y. The outcome-contingent contract $w : \mathcal{Y} \to \mathbb{R}_+$ from the baseline model is nested in this larger contract space by setting $w_t \equiv w$ for all t.

If the agent stops at time $\tau \geq 1$ under policy σ, then, with recall, she reports an outcome

$$\hat{Y}(\sigma, W) \in \operatorname*{argmax}_{s \in \{1,\dots,\tau\}} w_\tau(Y_s),$$

with ties broken in the principal's favor. If $\tau = 0$, both parties receive zero by convention. Let $\sigma_K^*(W)$ denote the selected best response under the contract W with the trial cap K, and define the principal's time-contingent second-best payoff by

$$\mathsf{SB}_P^\tau(K) := \sup_W u_P\big(W, \sigma_K^*(W)\big).$$

Because outcome-contingent contracts are nested within the larger class of time-contingent contracts, we have $\mathsf{SB}_P^\tau(K) \geq \mathsf{SB}_P(K)$ for all K. The next proposition shows that allowing payments to depend on τ restores monotonicity of the principal's payoff in the trial cap K, a comparative static that fails in the baseline model and is illustrated for outcome-only contracts in Figure 1.

Proposition 6 (Monotonicity of $\mathsf{SB}_P^\tau(K)$ in K)**.** *For every $K \geq 1$, we have $\mathsf{SB}_P^\tau(K+1) \geq \mathsf{SB}_P^\tau(K)$.*

The intuition is a nesting argument. When the cap increases from K to $K+1$, the principal can reproduce any cap-K time-contingent contract by setting the added period's payment rule to zero, $w_{K+1} \equiv 0$. Since all actions have strictly positive costs, the agent has no incentive to use that extra period to receive zero payment. The added trial is therefore effectively shut down, so every payoff attainable under cap K remains attainable under cap $K+1$.

This comparison clarifies why monotonicity fails in the baseline model. Increasing K expands the agent's hidden dynamic action space, much like adding actions in a static moral-hazard problem. Under outcome-only contracts, the principal cannot condition incentives on when the agent stops, so these additional dynamic incentive constraints can reduce her payoff. Time-contingent contracts restore monotonicity because they allow the principal to neutralize unwanted extra periods, making the larger-cap problem nest the smaller-cap problem.

Corollary 1 (Asymptotic guarantee under time-contingent contracts)**.** *Although the $\bar{\zeta}$-debt contract is time independent, it satisfies the same asymptotic guarantee relative to the larger class of time-contingent contracts. Formally, for every $K \geq 1$,*

$$\frac{U_P^{\text{debt}}(K;\bar{\zeta})}{\mathsf{SB}_P^{\tau}(K)} \geq 1-(1-\bar{q})^K.$$

Corollary 1 shows that the performance guarantee for the $\bar{\zeta}$-debt contract in Theorem 1 is robust to this larger contract space. The reason is that the proof of Theorem 1 actually benchmarks the $\bar{\zeta}$-debt payoff against the principal's first-best self-search value. Since this first-best value also upper-bounds what any time-contingent contract can deliver, the same argument yields Corollary 1.

6.2 Markov Correlated Sampling

We next relax the independent-sampling assumption. In many iterative workflows, the distribution of the next attempt depends on the best outcome obtained so far: a different best-so-far outcome can change both the likelihood and the size of future improvements. To capture this dependence while keeping the model tractable, we focus on the infinite-horizon case ($K = \infty$) and study a Markov model in which the distribution of the next draw depends on the current best outcome. We show that, under a natural monotonicity condition, an optimal contract is still a debt contract.

As in the baseline model, each time the agent samples action $i \in \mathcal{A}$, she incurs cost $c_i > 0$, and resampling is allowed. Relabel the raw outcome space as

$$\mathcal{Y} = \{0, 1, \ldots, m\}, \qquad 0 = v(0) < v(1) < \cdots < v(m). \tag{11}$$

After the first t trials, define $s_t := \max\{Y_1, \ldots, Y_t\}$, with the convention $s_0 = 0$. Because v is strictly increasing, s_t is the previously realized outcome with the highest principal value; we refer to it as the *incumbent*. If the agent chooses action $i \in \mathcal{A}$ when the current incumbent is s, a fresh draw $Z \sim P_i(\cdot \mid s)$ is generated, and the incumbent updates to $s \vee Z := \max\{s, Z\}$. The baseline independent-sampling model is nested as the special case $P_i(\cdot \mid s) \equiv p_i(\cdot)$ for all $s \in \mathcal{Y}$, so fresh draws are independent of the current incumbent.

Contracts remain outcome-contingent payment rules $w : \mathcal{Y} \to \mathbb{R}_+$. Upon stopping, the agent chooses among the raw outcomes realized in her history and reports one that maximizes her payment; ties among agent-optimal reports are broken in the principal's favor. The incumbent state also indexes the principal's first-best problem: in that benchmark, the principal directly controls sampling and keeps the best realized outcome. Let $J : \mathcal{Y} \to \mathbb{R}_+$ denote the principal's first-best continuation value in the infinite-horizon problem when the current incumbent is s. It is the unique bounded solution to

$$J(s) = \max\Big\{v(s),\ \max_{i\in\mathcal{A}}\Big[-c_i + \sum_{z\in\mathcal{Y}} P_i(z \mid s)\, J(s \vee z)\Big]\Big\}. \tag{12}$$

Theorem 2 (Optimality of debt contracts under correlated sampling ($K = \infty$))**.** *Consider the Markov correlated model with an infinite trial cap ($K = \infty$). Suppose that $J(s) \geq J(0)$ for every $s \in \mathcal{Y}$. Then the $J(0)$-debt contract*

$$w^{\mathsf{debt}}(y; J(0)) = \big(v(y) - J(0)\big)^+$$

is second-best optimal. Under an agent's best response, it implements a first-best sampling and stopping policy, gives the principal expected payoff $J(0)$, and leaves the agent with zero expected payoff.

We show in the proof that the shifted first-best value $J(s) - J(0)$ satisfies the agent's Bellman equation under the $J(0)$-debt contract. The argument compares the agent's Bellman equation with the first-best Bellman equation after subtracting the constant $J(0)$ from the first-best value function. Given the candidate value $J(s) - J(0)$, the continuation payoff from each action in the agent's Bellman equation is the corresponding first-best continuation payoff minus the same constant $J(0)$. Therefore the ranking of continuation actions is preserved, and the agent chooses first-best actions whenever she continues. The only remaining issue is the stopping term. In the shifted first-best equation, stopping gives $v(s)-J(0)$, whereas the debt contract gives $(v(s)-J(0))^+$. The assumption $J(s) \geq J(0)$ makes this difference harmless. If $v(s) \geq J(0)$, the two stopping payoffs coincide. If $v(s) < J(0)$, then stopping is strictly suboptimal in the first-best problem, because $J(s) \geq J(0) > v(s)$. Under the debt contract, the agent's stopping payoff is 0, while the continuation payoff from a first-best action is $J(s) - J(0) \geq 0$. Thus, continuation is also optimal for the agent, with principal-favor tie-breaking selecting continuation when needed. Therefore, replacing the negative shifted stopping payoff $v(s) - J(0)$ by the debt payoff 0 does not change the agent's choice that coincides with the first-best policy. Hence, the shifted value $J(s) - J(0)$ satisfies the agent's Bellman equation under the $J(0)$-debt contract. The agent therefore earns zero ex ante, while the principal obtains $J(0)$. Since $J(0)$ is the first-best total-surplus bound, no contract can give the principal more; thus the $J(0)$-debt contract is optimal.

A few comments are in order. First, in the independent-sampling benchmark, $P_i(\cdot \mid s) \equiv p_i(\cdot)$ for all s, the first-best value satisfies $J(0) = \bar{\zeta}$, where $\bar{\zeta}$ is the maximal Weitzman reservation value defined in Equation (8). Thus $w^{\mathsf{debt}}(y; J(0))$ coincides with the $\bar{\zeta}$-debt contract in Theorem 1, and Theorem 2 recovers the infinite-horizon result from the baseline model. Second, unlike the independent-sampling benchmark, *this result is not a no-switching result.* Under the $\bar{\zeta}$-debt contract in the baseline model, the agent repeatedly samples a maximal-index action until the cutoff is reached. With correlated sampling, however, the agent may switch actions under the $J(0)$-debt contract. The reason is that the contract works by aligning the agent's Bellman problem with the principal's first-best Bellman problem when the principal directly controls sampling. In the Markov model, the first-best rule can be state dependent because the continuation payoff from action i depends on the current incumbent s. Thus the agent may switch actions as the incumbent

changes, even though her policy remains first-best.

The condition $J(s) \geq J(0)$ is stated in terms of the endogenous first-best value function. The next result gives a sufficient condition, expressed directly in terms of the sampling distributions $P_i(\cdot \mid s)$.

Proposition 7 (A first-order stochastic dominance sufficient condition for $J(s) \geq J(0)$)**.** *For each action $i \in \mathcal{A}$ and incumbent $s \in \mathcal{Y}$, let $Z_{i,s} \sim P_i(\cdot \mid s)$ and define $X_i(s) := s \vee Z_{i,s}$. Thus $X_i(s)$ is the next incumbent if action i is sampled when the current incumbent is s. Suppose that, for every action $i \in \mathcal{A}$, every pair of incumbents $s' \geq s$, and every outcome $y \in \mathcal{Y}$,*

$$\Pr(X_i(s') \geq y) \;\geq\; \Pr(X_i(s) \geq y). \tag{FOSD}$$

Then, the first-best value function J is nondecreasing in the incumbent, and hence $J(s) \geq J(0)$ for every $s \in \mathcal{Y}$.

Condition (FOSD) says that, holding the action fixed, starting from a higher incumbent cannot make the next best-so-far outcome worse in the sense of first-order stochastic dominance. Equivalently, for every outcome y, the probability that the next incumbent reaches at least y must be weakly higher when the current incumbent is higher. Importantly, the condition is imposed on the next incumbent $X_i(s) = s \vee Z_{i,s}$, not directly on the fresh draw $Z_{i,s}$. Thus, the fresh sample distribution may depend on the incumbent in flexible ways; what matters is that, after updating the state to the best-so-far outcome, a higher incumbent does not worsen the distribution of the next incumbent.

The condition is satisfied in several natural cases. It holds in the independent-sampling benchmark, where $P_i(\cdot \mid s) \equiv p_i(\cdot)$, because increasing s mechanically raises $s \vee Z$. More generally, the condition allows for diminishing returns to search. Large improvements may become less likely when the incumbent is already high, but the overall distribution of the next incumbent must still shift weakly upward. What the condition rules out are cases in which a higher incumbent makes future search so much less productive that the next-incumbent distribution becomes worse. The following example gives a simple local-refinement process satisfying the condition.

Example 1 (Local-refinement sampling)**.** *Fix an action $i \in \mathcal{A}$. Let D_i be an integer-valued random shock with distribution q_i, supported on a finite subset of $\mathbb{Z}$. For each incumbent $s \in \mathcal{Y}$, define the raw draw under action i by*

$$Z_i(s) := \min\{m, \max\{0, s + D_i\}\}.$$

The truncation keeps $Z_i(s)$ in $\mathcal{Y} = \{0, 1, \ldots, m\}$. The induced next incumbent is

$$X_i(s) := s \vee Z_i(s) = \min\{m,\ s + D_i^+\}, \qquad D_i^+ := \max\{D_i, 0\}.$$

Thus $X_i(s)$ is increasing in s for every realization of D_i, so (FOSD) holds.

This model captures local refinement of the current best candidate. The shock D_i is the local change generated by action i: negative shocks produce worse raw candidates, positive shocks produce better raw candidates, and the truncation keeps outcomes inside $\mathcal{Y}$. Because only the best-so-far outcome is retained, negative shocks cannot lower the incumbent, while positive shocks translate a higher current incumbent into a weakly higher next incumbent.

6.3 Multiple Agents

We finally extend the model to settings in which the principal can draw sequentially on multiple agents. This captures environments in which the principal may solicit outputs from several workers, teams, or systems, but ultimately adopts only one reported outcome. Additional agents create more search opportunities, while the principal's payoff depends on the best report she adopts rather than on the sum of all reports.

Specifically, there is a finite set of agents $\mathcal{G} := \{1, \dots, G\}$. Each agent $g \in \mathcal{G}$ has a finite action set $\mathcal{A}_g := \{1, \dots, n_g\}$. Agent g can take at most $K_g \geq 1$ trials, where K_g is exogenously given. The principal can contract with agents sequentially, with each agent contracted at most once. A contract for agent g is a payment rule $w_g : \mathcal{Y} \to \mathbb{R}_+$ that maps agent g's reported outcome to her payment. If the principal contracts with agent g, the agent privately solves the same adaptive experimentation problem as in the single-agent model: at each inner trial $k \in \{1, \dots, K_g\}$, she chooses an action $a_{gk} \in \mathcal{A}_g$, pays cost $c_{ga_{gk}}$, and observes an outcome $Y_{gk} \in \mathcal{Y}$ drawn from a known distribution $p_{ga_{gk}} \in \Delta(\mathcal{Y})$. Repeated samples from the same action are i.i.d.; conditional on the chosen action sequences, outcomes are independent across trials and across agents. Let $h_k^g := (a_{g1}, Y_{g1}, \dots, a_{gk}, Y_{gk})$ denote agent g's inner history after k completed trials, with $h_0^g := \emptyset$. As in the baseline model, a policy σ_g specifies, after each history h_k^g with $k < K_g$, whether agent g stops or continues and, if she continues, which action she takes next. Let $\tau_g \in \{0, 1, \dots, K_g\}$ be the induced stopping time. If $\tau_g \geq 1$, agent g reports an outcome from her own history that maximizes her payment:

$$\hat{Y}_g(\sigma_g, w_g) \in \operatorname*{argmax}_{k \in \{1, \dots, \tau_g\}} w_g(Y_{gk}),$$

with ties broken in the principal's favor. If $\tau_g = 0$, we write $\hat{Y}_g = \varnothing$, and both payment and cost are zero by convention. We extend both w_g and v to $\mathcal{Y} \cup \{\varnothing\}$ by setting $w_g(\varnothing) = v(\varnothing) := 0$. If agent g is contracted and follows policy σ_g under contract w_g, her expected utility is

$$u_{A,g}(w_g, \sigma_g) := \mathbb{E}\left[w_g(\hat{Y}_g(\sigma_g, w_g)) - \sum_{k=1}^{\tau_g} c_{ga_{gk}} \right].$$

If agent g is never contracted, her utility is 0.

After t agents have been contracted, the outer history is $H_t := (g_1, \hat{Y}_{g_1}, \dots, g_t, \hat{Y}_{g_t})$, where $g_1, \dots, g_t$ are distinct contracted agents. Let $\mathcal{S}_t := \{g_1, \dots, g_t\}$, with $H_0 = \emptyset$ and $\mathcal{S}_0 = \emptyset$. If the

principal stops at history H_t, she adopts the best report received so far. Her realized payoff is

$$\max_{g\in\mathcal{S}_t} v(\hat{Y}_g) - \sum_{g\in\mathcal{S}_t} w_g(\hat{Y}_g), \tag{13}$$

with payoff 0 when $\mathcal{S}_t = \emptyset$. Thus values do not accumulate across reports: only the best report enters gross value, while payments are made to all contracted agents. The principal chooses an outer policy π that specifies, after each outer history H_t, whether to stop or to contract with another agent in $\mathcal{G}\setminus\mathcal{S}_t$, and, if so, which agent to contract with next.[3] Given $(\pi,(w_g)_{g\in\mathcal{G}})$, each contracted agent g chooses a best response $\sigma_g^*(w_g)$, with ties broken in the principal's favor. Let $u_P(\pi,(w_g)_{g\in\mathcal{G}})$ denote the principal's expected terminal payoff when contracted agents best respond, with terminal payoffs computed from Equation (13). Define

$$\mathsf{SB}_P(\mathbf{K}) := \max_{\pi,\,(w_g)_{g\in\mathcal{G}}} u_P(\pi,(w_g)_{g\in\mathcal{G}}), \qquad \mathbf{K} := (K_1,\dots,K_G),$$

as the principal's multi-agent second-best payoff. Let $\mathsf{FB}_P(\mathbf{K})$ denote the corresponding first-best benchmark in which the principal directly controls all agents' experimentation subject to the same caps $\mathbf{K}$, observes all realized outcomes, keeps the best realized outcome, and bears the sampling costs.

We now define a contracting strategy based on a sequence of debt contracts and show that it is asymptotically optimal. For every agent–action pair (g,i), let ζ_{gi} denote the Weitzman reservation value, defined as the unique solution to

$$\mathbb{E}_{Y\sim p_{gi}}\big[(v(Y)-\zeta_{gi})^+\big] = c_{gi}. \tag{14}$$

For each agent g, let $\bar{\zeta}_g := \max_{i\in\mathcal{A}_g}\zeta_{gi}$ be her highest reservation value, and let $\bar{i}_g \in \operatorname{argmax}_{i\in\mathcal{A}_g}\zeta_{gi}$ be an action that attains it. Without loss of generality, relabel agents so that $\bar{\zeta}_1 \ge \bar{\zeta}_2 \ge \cdots \ge \bar{\zeta}_G$. Let $\mathcal{G}^* := \{g\in\mathcal{G} : \bar{\zeta}_g = \bar{\zeta}_1\}$ and set $\bar{\zeta}_{G+1} := 0$. Assume $\bar{\zeta}_1 > 0$; otherwise, all agent-action pairs have nonpositive first-best indices, and it is optimal for the principal to contract with nobody.

Definition 1 (Sequential $\bar{\zeta}$-debt contracting strategy)**.** *The sequential $\bar{\zeta}$-debt contracting strategy consists of the outer policy and the agent-specific debt contracts $(\pi^{\mathsf{seq\text{-}debt}},(w_g^{\mathsf{seq\text{-}debt}})_{g\in\mathcal{G}})$, defined as follows.*

Contracts. *For each agent $g\in\mathcal{G}$, the contract is the $\bar{\zeta}_g$-debt contract*

$$w_g^{\mathsf{seq\text{-}debt}}(y) := (v(y)-\bar{\zeta}_g)^+, \qquad y\in\mathcal{Y}. \tag{15}$$

Contracting sequence. *The outer policy $\pi^{\mathsf{seq\text{-}debt}}$ approaches agents in decreasing order of their maximal indices, from agent 1 to agent G. After the first t agents have been contracted, with outer history $H_t = (1,\hat{Y}_1,\dots,t,\hat{Y}_t)$, the principal stops if and only if $X_t := \max_{r\in\mathcal{S}_t} v(\hat{Y}_r) = \max_{1\le r\le t} v(\hat{Y}_r) \ge \bar{\zeta}_{t+1}$.*

[3] Because the principal cares only about the best reported outcome, it is never optimal for the principal to contract with multiple agents simultaneously.

Let $U_P^{\textsf{seq-debt}}(\mathbf{K}) := u_P(\pi^{\textsf{seq-debt}}, (w_g^{\textsf{seq-debt}})_{g\in\mathcal{G}})$ denote the principal's expected payoff under the sequential $\bar{\zeta}$-debt contracting strategy in Definition 1. For each $g \in \mathcal{G}$, define the single-trial probability that agent g's maximal-index action reaches her maximal Weitzman reservation value in a single sample:

$$q_g := \Pr_{Y\sim p_{g,\bar{i}_g}}\{v(Y) \geq \bar{\zeta}_g\}. \tag{16}$$

Theorem 3 (Asymptotic Optimality of Sequential $\bar{\zeta}$-Debt Contracts)**.** *Under the sequential $\bar{\zeta}$-debt contracting strategy in Definition 1, for every cap vector $\mathbf{K}$,*

$$\frac{U_P^{\textsf{seq-debt}}(\mathbf{K})}{\textsf{SB}_P(\mathbf{K})} \geq \frac{1}{\bar{\zeta}_1}\sum_{g=1}^{G}\bar{\zeta}_g\Big(1-(1-q_g)^{K_g}\Big)\prod_{\ell<g}(1-q_\ell)^{K_\ell} \tag{17}$$

$$\geq 1-\prod_{g\in\mathcal{G}^*}(1-q_g)^{K_g}, \tag{18}$$

where $\prod_{\ell<1}(\cdot) := 1$. Moreover, if $g_0 := \min\{g \in \mathcal{G} : K_g = \infty\}$ exists, then

$$U_P^{\textsf{seq-debt}}(\mathbf{K}) \geq \bar{\zeta}_{g_0}, \qquad \frac{U_P^{\textsf{seq-debt}}(\mathbf{K})}{\textsf{SB}_P(\mathbf{K})} \geq \frac{\bar{\zeta}_{g_0}}{\bar{\zeta}_1}. \tag{19}$$

In particular, if some highest-index agent has an infinite cap, that is, if $g_0 \in \mathcal{G}^*$, then $U_P^{\textsf{seq-debt}}(\mathbf{K}) = \textsf{SB}_P(\mathbf{K}) = \bar{\zeta}_1$.

Theorem 3 extends the single-agent guarantee in Theorem 1 to the multi-agent setting. The proof follows the same upper-bound/lower-bound structure as in the single-agent case. The second-best benchmark $\textsf{SB}_P(\mathbf{K})$ is upper-bounded by the infinite-horizon first-best value $\bar{\zeta}_1$, while the payoff from sequential debt is lower-bounded by aggregating threshold-reaching events across agents. The term $\bar{\zeta}_g(1-(1-q_g)^{K_g})\prod_{\ell<g}(1-q_\ell)^{K_\ell}$ corresponds to histories in which agents $1,\dots,g-1$ fail to reach their own debt thresholds and agent g would reach hers if approached. On such histories, the principal's payoff is at least $\bar{\zeta}_g$: either she stops earlier because an intermediate report already exceeds the next cutoff, which is at least $\bar{\zeta}_g$, or she reaches agent g, who crosses her own threshold and delivers payoff $\bar{\zeta}_g$. If some agent has an infinite cap, let g_0 be the first such agent in the index-ordered sequence. Under the sequential $\bar{\zeta}$-debt strategy, no agent after g_0 is contracted, and the principal's payoff is guaranteed to be at least $\bar{\zeta}_{g_0}$.

The simpler bound in (18) keeps only the highest-index agents in $\mathcal{G}^*$. This highlights the source of asymptotic optimality: the second-best payoff is bounded above by $\bar{\zeta}_1$, and whenever any highest-index agent reaches her threshold, sequential debt secures this upper-bound payoff level. Since $q_g > 0$ for every $g \in \mathcal{G}^*$,[4] the probability of this event, $1-\prod_{g\in\mathcal{G}^*}(1-q_g)^{K_g}$, converges to 1 as the cumulative experimentation cap of the highest-index agents grows. In particular, if at least one highest-index agent has an infinite cap, the sequential $\bar{\zeta}$-debt strategy is exactly second-best

[4]Indeed, $c_{g\bar{i}_g} > 0$ and $\mathbb{E}[(v(Y)-\bar{\zeta}_g)^+] = c_{g\bar{i}_g}$, so $q_g = \Pr\{v(Y) \geq \bar{\zeta}_g\} > 0$.

optimal.

Finally, the bound in (17) is conservative. The proof compares sequential debt to the stronger upper bound $\bar{\zeta}_1$, the first-best value with unbounded caps, rather than only to the finite-cap second-best payoff $\mathsf{SB}_P(\mathbf{K})$. It also counts only histories in which some agent reaches her own debt threshold, ignoring histories where an agent falls short of $\bar{\zeta}_g$ but still produces an intermediate report high enough to make the principal stop. The actual performance of sequential debt can therefore exceed the stated guarantee, especially when trial caps are small.

7 Conclusion

We show that when hidden effort becomes sequential, the incentive problem changes qualitatively. The principal is no longer only inducing one costly action; she must discipline a private search process in which the agent can try, revise, resample, and decide when to stop before reporting a final outcome. The main takeaway is that simple threshold-based payments can be powerful in such environments. A calibrated debt contract suppresses rewards for intermediate outcomes and focuses incentives on reaching a sufficiently high final outcome. This structure limits the dynamic distortions created by hidden continuation opportunities and becomes increasingly effective as the trial cap grows. More broadly, the results suggest that in AI-enabled and other experimentation-intensive settings, outcome-based contracts should account for how repeated hidden attempts reshape the agent's incentives.

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Appendix

Appendix A provides the detailed algorithm for computing the optimal finite-K debt contract in Proposition 3. Appendix B collects the proofs of all results.

A Auxiliary Results: Computing the Optimal Debt Contract

This appendix presents the algorithm underlying Proposition 3; the formal proof is given in Section B.2. Relabel outcomes so that $0 = v(0) \leq v(1) \leq \cdots \leq v(m)$, where 0 is a fictitious null outcome with $v(0) := 0$. If several outcomes have the same value, break ties arbitrarily; debt payoffs depend only on $v(y)$. Let

$$\mathcal{Y}_0 := \{0\} \cup \mathcal{Y} = \{0, 1, \ldots, m\}.$$

The algorithm exploits a special feature of debt contracts. Once the face value x is fixed, the payoff-relevant information in the history is the best realized outcome so far, which we call the *incumbent*. Because outcomes are ordered by value, the incumbent can be represented by the largest realized outcome label. (Under principal-favor tie-breaking, if several recalled outcomes give the agent the same payment, the agent reports the one that gives the principal the highest value.) Thus the incumbent summarizes the payoff-relevant history for *both* parties. Hence the dynamic state is (k, s), where k is the number of remaining trials and $s \in \mathcal{Y}_0$ is the incumbent outcome label, with $s = 0$ before any trial.

For each state (k, s), let $(A_k(s; x), P_k(s; x))$ denote the equilibrium continuation payoff pair under principal-favor tie-breaking. At $k = 0$,

$$A_0(s; x) = (v(s) - x)^+, \qquad P_0(s; x) = \min\{v(s), x\}.$$

For $k \geq 1$, the agent either stops immediately or chooses an action i. If action i is chosen, the continuation pair is

$$\left(-c_i + \sum_{y \in \mathcal{Y}} p_i(y)\, A_{k-1}(\max\{s, y\}; x),\ \sum_{y \in \mathcal{Y}} p_i(y)\, P_{k-1}(\max\{s, y\}; x) \right),$$

and the equilibrium pair at (k, s) is the lexicographic maximum (maximize agent payoff, tie-break by principal payoff) over the stop option and all action candidates. The game starts from $(K, 0)$, so $U_P^{\mathsf{debt}}(K; x) = P_K(0; x)$.

The key structural observation is that, viewed as a function of the face value $x \in [0, v_{\max}]$ where $v_{\max} := \max_{y \in \mathcal{Y}} v(y)$, every continuation value pair $(A_k(s; \cdot), P_k(s; \cdot))$ is piecewise affine with finitely many breakpoints. A breakpoint representation of a payoff pair $x \mapsto (A(x), P(x))$ consists of a finite breakpoint list, affine formulas for $(A(x), P(x))$ on each open interval determined by the list, and point values at the breakpoints. Algorithm 1 below maintains an explicit breakpoint

representation of these functions and computes the optimal face value by sweeping through the breakpoint list of $P_K(0;\cdot)$. The supporting subroutines TryCandidate, LexMaxEnvelope, and StopPair construct, respectively, the candidate from sampling action i, the pointwise lexicographic envelope across candidates, and the breakpoint representation of the stop option.

Algorithm 1 DebtOpt(K): exact optimal debt face value for finite K

Input: values $v(y)$, distributions p_i, costs c_i, and trial cap K.
Output: an optimal debt face value $x^*_{\mathsf{debt}}(K)$.
$v_{\max} \leftarrow \max_{y\in\mathcal{Y}} v(y)$.
if $v_{\max} = 0$ **then**
    return 0.
end if
for each $s \in \mathcal{Y}_0$ **do**
    $\mathcal{R}_0(s) \leftarrow \textsc{StopPair}(s, v_{\max})$.
end for
for $k = 1, \dots, K$ **do**
    for each $s \in \mathcal{Y}_0$ **do**
        Initialize an empty candidate list $\mathcal{C} \leftarrow \emptyset$.
        Append $\textsc{StopPair}(s, v_{\max})$ to $\mathcal{C}$.
        for each $i \in \mathcal{A}$ **do**
            Append $\textsc{TryCandidate}(i, k, s, \mathcal{R}_{k-1}, v_{\max})$ to $\mathcal{C}$.
        end for
        $\mathcal{R}_k(s) \leftarrow \textsc{LexMaxEnvelope}(\mathcal{C}, v_{\max})$.
    end for
end for
Let $B_K(0)$ be the breakpoint list stored in $\mathcal{R}_K(0)$.
Evaluate $P_K(0;x)$ for every $x \in B_K(0)$.
return any $x \in \operatorname{argmax}_{z\in B_K(0)} P_K(0;z)$.

B Proof

B.1 Proof of Section 4

Proof of Lemma 1. Fix a contract w. We suppress the superscript w and write $V_k(\cdot)$ and $\Psi_k(\cdot, i)$.

Step 1: V_k is nondecreasing, 1-Lipschitz, and convex.

We prove the three properties by induction on k.

Base case ($k = 0$). $V_0(b) = b$ is nondecreasing, 1-Lipschitz, and convex on $\mathbb{R}$.

Inductive step. Fix $k \geq 1$ and assume V_{k-1} is nondecreasing, 1-Lipschitz, and convex. For each action $i \in \mathcal{A}$, define

$$F_i(b) := -c_i + \mathbb{E}_{Y\sim p_i}\left[V_{k-1}(\max\{b, w(Y)\})\right].$$

Algorithm 2 TryCandidate: candidate pair from sampling action i

Input: action i, layer k, state $s \in \mathcal{Y}_0$, previous representations $\mathcal{R}_{k-1}(\cdot)$, endpoint $v_{\max}$.
Output: a breakpoint representation of the action-i candidate pair.
for each $y \in \mathcal{Y}$ **do**
    $s_y \leftarrow \max\{s, y\}$.
end for
$B \leftarrow \text{UniqueSort}(\bigcup_{y\in\mathcal{Y}} B_{k-1}(s_y))$.
for each open interval I determined by B **do**
    Retrieve affine formulas $A_{k-1}(s_y; x)$ and $P_{k-1}(s_y; x)$ valid on I.
    Store on I:

$$A(x) = -c_i + \sum_{y\in\mathcal{Y}} p_i(y) A_{k-1}(s_y; x), \qquad P(x) = \sum_{y\in\mathcal{Y}} p_i(y) P_{k-1}(s_y; x).$$

end for
for each breakpoint $b \in B$ **do**
    Store point values

$$A(b) = -c_i + \sum_{y\in\mathcal{Y}} p_i(y) A_{k-1}(s_y; b), \qquad P(b) = \sum_{y\in\mathcal{Y}} p_i(y) P_{k-1}(s_y; b).$$

end for
return the resulting representation.

Monotonicity of F_i. If $b' \geq b$, then $\max\{b', x\} \geq \max\{b, x\}$ for all x and, since V_{k-1} is nondecreasing,

$$V_{k-1}(\max\{b', w(Y)\}) \geq V_{k-1}(\max\{b, w(Y)\}) \quad \text{a.s.}$$

Taking expectations gives $F_i(b') \geq F_i(b)$, so F_i is nondecreasing.

1-*Lipschitzness of F_i.* Fix $b', b \in \mathbb{R}$ and set

$$\delta(Y) := \max\{b', w(Y)\} - \max\{b, w(Y)\}.$$

Then $|\delta(Y)| \leq |b' - b|$ almost surely. By 1-Lipschitzness of V_{k-1},

$$|V_{k-1}(\max\{b', w(Y)\}) - V_{k-1}(\max\{b, w(Y)\})| \leq |\delta(Y)| \leq |b' - b|.$$

Taking expectations yields $|F_i(b') - F_i(b)| \leq |b' - b|$, so F_i is 1-Lipschitz.

Convexity of F_i. The map $b \mapsto \max\{b, w(Y)\}$ is convex for each realized Y, and since V_{k-1} is convex and nondecreasing, the composition $b \mapsto V_{k-1}(\max\{b, w(Y)\})$ is convex. Expectations preserve convexity, hence F_i is convex.

Let $M(b) := \max_{i\in\mathcal{A}} F_i(b)$. We first claim that M is nondecreasing and convex. The monotonicity of M is immediate. Convexity follows because M is a finite pointwise maximum of convex functions. We next claim that M is 1-Lipschitz. For this, fix $b' \geq b$ and choose $i' \in \operatorname{argmax}_{i\in\mathcal{A}} F_i(b')$. Then

$$M(b') - M(b) = F_{i'}(b') - \max_{i\in\mathcal{A}} F_i(b) \leq F_{i'}(b') - F_{i'}(b) \leq b' - b.$$

Interchanging b and b' gives $|M(b') - M(b)| \leq |b' - b|$, so M is 1-Lipschitz. The same argument

Algorithm 3 LexMaxEnvelope: pointwise lexicographic envelope

Input: candidate representations $\mathcal{C} = (\mathcal{C}^1, \dots, \mathcal{C}^{n+1})$, all on $[0, v_{\max}]$.
Here $n = |\mathcal{A}|$; in Algorithm 1, these are the stop candidate and one action candidate for each action.
Output: representation of $\max_{\succeq_{\text{lex}}}\{\mathcal{C}^1, \dots, \mathcal{C}^{n+1}\}$.
$B^0 \leftarrow \text{UniqueSort}(\bigcup_{r=1}^{n+1} B^r)$.
Initialize an empty global breakpoint list B and empty interval formulas.
for each open interval $I = (a, b)$ determined by B^0 **do**
    Retrieve affine formulas $A^r(x)$ and $P^r(x)$ for every $r = 1, \dots, n+1$.
    $T \leftarrow \{a, b\}$.
    for each pair $1 \le r < \ell \le n+1$ **do**
        If $A^r(x) - A^\ell(x)$ is not identically zero on I and has a root in (a, b), add it to T.
        If $P^r(x) - P^\ell(x)$ is not identically zero on I and has a root in (a, b), add it to T.
    end for
    Sort $T = \{t_0 < t_1 < \cdots < t_q\}$ and add all points of T to B.
    for $h = 1, \dots, q$ **do**
        Choose any $\xi \in (t_{h-1}, t_h)$.
        Choose r^* such that $(A^{r^*}(\xi), P^{r^*}(\xi)) \succeq_{\text{lex}} (A^r(\xi), P^r(\xi))$ for every $r = 1, \dots, n+1$.
        Store candidate r^*'s affine formulas on (t_{h-1}, t_h).
    end for
end for
Sort and deduplicate the global breakpoint list B.
for each breakpoint $b \in B$ **do**
    Evaluate every candidate $r = 1, \dots, n+1$ pointwise at b.
    Store the pointwise lexicographic maximum value at b.
end for
return the breakpoint representation of the selected pair.

Algorithm 4 StopPair: breakpoint representation of the stop option

Input: state $s \in \mathcal{Y}_0$, endpoint $v_{\max}$.
Output: breakpoint representation of $((v(s) - x)^+, \min\{v(s), x\})$ on $[0, v_{\max}]$.
$B \leftarrow \text{UniqueSort}(\{0, v(s), v_{\max}\})$.
for each open interval I determined by B **do**
    Store affine formulas for $(v(s) - x)^+$ and $\min\{v(s), x\}$ on I.
end for
for each breakpoint $b \in B$ **do**
    Store point values $(v(s) - b)^+$ and $\min\{v(s), b\}$.
end for
return the resulting representation.

applies to the finite maximum $\max\{b, M(b)\}$, because both b and $M(b)$ are nondecreasing and 1-Lipschitz. Hence $V_k(b) = \max\{b, M(b)\}$ is nondecreasing and 1-Lipschitz; it is also convex as the maximum of two convex functions. This completes the induction.

Step 2: $\Psi_k(\cdot, i)$ is continuous, convex, nonincreasing, and has a unique zero.

Recall $\Psi_k(b, i) = F_i(b) - b$. Since F_i is 1-Lipschitz, $\Psi_k(\cdot, i)$ is continuous. Since F_i is convex, $\Psi_k(\cdot, i) = F_i(\cdot) - (\cdot)$ is convex.

Moreover, for $b' > b$,

$$\Psi_k(b', i) - \Psi_k(b, i) = \big(F_i(b') - F_i(b)\big) - (b' - b) \le (b' - b) - (b' - b) = 0,$$

where we used that F_i is 1-Lipschitz (hence $F_i(b') - F_i(b) \le b' - b$). Thus $\Psi_k(\cdot, i)$ is nonincreasing on $\mathbb{R}$.

Existence of a solution. Let

$$\bar{w} := \max\{w(y) : \exists j \in \mathcal{A} \text{ with } p_{jy} > 0\}.$$

We first claim that for every $t \in \{0, 1, \dots, K\}$ and every $b \ge \bar{w}$, we have $V_t(b) = b$. This is by induction on t: for $t = 0$ it is immediate. If $V_{t-1}(b) = b$ for $b \ge \bar{w}$, then for such b and any action j, $\max\{b, w(Y)\} = b$ almost surely under $Y \sim p_j$, so

$$-c_j + \mathbb{E}[V_{t-1}(\max\{b, w(Y)\})] = -c_j + \mathbb{E}[V_{t-1}(b)] = -c_j + b < b.$$

Hence $V_t(b) = \max\{b, \max_j(\cdots)\} = b$.

Now fix i and $k \ge 1$. For any $b \ge \bar{w}$, we have $\max\{b, w(Y)\} = b$ almost surely under $Y \sim p_i$ and $V_{k-1}(b) = b$, therefore

$$\Psi_k(b, i) = -c_i + \mathbb{E}[V_{k-1}(\max\{b, w(Y)\})] - b = -c_i + V_{k-1}(b) - b = -c_i < 0.$$

On the other hand, as $b \to -\infty$, $\max\{b, w(Y)\} = w(Y)$ almost surely (since $w(Y)$ is bounded below), so

$$\Psi_k(b, i) = -c_i + \mathbb{E}[V_{k-1}(w(Y))] - b \longrightarrow +\infty.$$

By continuity, there exists at least one z with $\Psi_k(z, i) = 0$, i.e., (4) holds.

Uniqueness of the solution. Suppose for contradiction that there exist $z_1 < z_2$ such that $\Psi_k(z_1, i) = \Psi_k(z_2, i) = 0$. Because $z_2 < \bar{w}$ (since $\Psi_k(b, i) < 0$ for all $b \ge \bar{w}$ and $\Psi_k(\cdot, i)$ is nonincreasing), we can write $z_2 = \lambda z_1 + (1 - \lambda)\bar{w}$ for some $\lambda \in (0, 1)$. By convexity of $\Psi_k(\cdot, i)$,

$$0 = \Psi_k(z_2, i) \le \lambda \Psi_k(z_1, i) + (1 - \lambda)\Psi_k(\bar{w}, i) = (1 - \lambda)(-c_i) < 0,$$

a contradiction. Hence the solution is unique. Since $\Psi_k(b, i) < 0$ for all $b \ge \bar{w}$, the unique solution satisfies $z_k(i) < \bar{w}$. □

Proof of Proposition 1. Fix a contract w and suppress the superscript w for readability. From the Bellman recursion (2), for any $k \geq 1$ we can rewrite

$$V_k(b) = \max\Big\{ b,\ \max_{i\in\mathcal{A}} \Big(-c_i + \mathbb{E}[V_{k-1}(\max\{b, w(Y)\})]\Big)\Big\} = \max\Big\{ b,\ \max_{i\in\mathcal{A}} \big(b + \Psi_k(b,i)\big)\Big\}.$$

Therefore stopping is weakly optimal at state (k, b) if and only if

$$b \geq \max_{i\in\mathcal{A}} \big(b + \Psi_k(b,i)\big) \quad \Longleftrightarrow \quad \max_{i\in\mathcal{A}} \Psi_k(b,i) \leq 0.$$

Moreover, conditional on continuing, an optimal action choice is any

$$i^*(k,b) \in \arg\max_{i\in\mathcal{A}} \Psi_k(b,i),$$

since b is common across actions.

Now invoke Lemma 1(ii)–(iii): for each i and k the function $b \mapsto \Psi_k(b,i)$ is nonincreasing and admits a unique root $z_k(i)$ satisfying $\Psi_k(z_k(i), i) = 0$. Hence $\Psi_k(b,i) \leq 0$ if and only if $b \geq z_k(i)$, and $\Psi_k(b,i) > 0$ if and only if $b < z_k(i)$. Let $\bar{z}_k := \max_{i\in\mathcal{A}} z_k(i)$.

If $b > \bar{z}_k$, then $b > z_k(i)$ for every i. By nonincreasingness and uniqueness of the root, $\Psi_k(b,i) < 0$ for every i, hence $\max_i \Psi_k(b,i) < 0$ and stopping is strictly optimal.

If $b < \bar{z}_k$, choose any i with $z_k(i) = \bar{z}_k$. Then $b < z_k(i)$, so $\Psi_k(b,i) > 0$ and therefore $\max_i \Psi_k(b,i) > 0$, implying that continuing is strictly optimal. Conditional on continuing, the optimal action is any maximizer of $\Psi_k(b,i)$, with principal-favor tie-breaking among action ties.

Finally, if $b = \bar{z}_k$, then $\Psi_k(b,i) \leq 0$ for all i, with equality for any action attaining $\bar{z}_k$. These equality actions are exactly the actions in $\operatorname{argmax}_{i\in\mathcal{A}} \Psi_k(b,i)$. Thus, the agent is indifferent between stopping and continuing with an action in this argmax set, and principal-favor tie-breaking selects among these agent-optimal choices. □

Proof of Proposition 2. Fix a contract w and suppress the superscript w throughout.

Proof of (i).

We first note a simple feasible-policy inclusion. For any $k \geq 1$ and any current best payment $b \in \mathbb{R}_+$, every policy feasible with $k-1$ remaining trials is also feasible with k remaining trials: the agent can follow that policy and simply stop before using the additional trial. Since $V_k(b)$ maximizes over a weakly larger feasible set of policies than $V_{k-1}(b)$, we have

$$V_k(b) \geq V_{k-1}(b), \qquad k = 1, \ldots, K,\ b \in \mathbb{R}_+. \tag{B-1}$$

We next use this monotonicity to prove that $z_k(i)$ is weakly increasing in k.

Preliminary: $\Psi_k(b,i)$ is pointwise nondecreasing in k. Fix an action $i \in \mathcal{A}$, a stage $k \in \{1, \ldots, K-1\}$, and $b \in \mathbb{R}_+$. By definition,

$$\Psi_{k+1}(b,i) = -c_i + \mathbb{E}_{Y\sim p_i}\Big[V_k\big(\max\{b, w(Y)\}\big)\Big] - b, \qquad \Psi_k(b,i) = -c_i + \mathbb{E}_{Y\sim p_i}\Big[V_{k-1}\big(\max\{b, w(Y)\}\big)\Big] - b.$$

Using (B-1) with $t = \max\{b, w(Y)\} \in \mathbb{R}_+$ (pointwise in Y) yields $V_k(\max\{b, w(Y)\}) \geq V_{k-1}(\max\{b, w(Y)\})$ almost surely, hence $\Psi_{k+1}(b, i) \geq \Psi_k(b, i)$ after taking expectations.

Conclusion: $z_k(i)$ is nondecreasing in k. Fix $i \in \mathcal{A}$ and $k \in \{1, \ldots, K-1\}$. Let $z_k(i)$ denote the unique solution to $\Psi_k(z, i) = 0$ and $z_{k+1}(i)$ the unique solution to $\Psi_{k+1}(z, i) = 0$. By the previous preliminary result,

$$\Psi_{k+1}(z_k(i), i) \;\geq\; \Psi_k(z_k(i), i) \;=\; 0.$$

By Lemma 1(ii), the function $b \mapsto \Psi_{k+1}(b, i)$ is nonincreasing, and by Lemma 1(iii) it has a unique zero at $z_{k+1}(i)$. If we had $z_{k+1}(i) < z_k(i)$, then nonincrease would imply

$$\Psi_{k+1}(z_k(i), i) \;\leq\; \Psi_{k+1}(z_{k+1}(i), i) \;=\; 0,$$

which contradicts $\Psi_{k+1}(z_k(i), i) \geq 0$ unless equality holds; but equality would force $\Psi_{k+1}(z_k(i), i) = 0$, and then uniqueness of the zero would imply $z_k(i) = z_{k+1}(i)$, a contradiction. Therefore $z_{k+1}(i) \geq z_k(i)$, proving (i).

Proof of (ii)(a).

Fix a contract w and suppose $K = \infty$. Throughout, suppress the superscript w.

Step 1: Stationary Bellman equation.

With an infinite budget, the agent's continuation value depends only on the current best payment b. Let $V_\infty(b)$ denote the agent's optimal expected utility (payment minus future costs) starting from best-so-far b. Then V_∞ satisfies the stationary Bellman equation

$$V_\infty(b) = \max\Big\{ b,\ \max_{i \in \mathcal{A}} \Big(-c_i + \mathbb{E}_{Y \sim p_i}\big[V_\infty(\max\{b, w(Y)\})\big] \Big) \Big\}. \tag{B-2}$$

Moreover, since $\mathcal{Y}$ is finite and $w(\cdot) \geq 0$, the payment is uniformly bounded by $\bar{w} := \max_{y \in \mathcal{Y}} w(y)$, hence $0 \leq V_\infty(b) \leq \bar{w}$ for all $b \in \mathbb{R}_+$.

Step 2: Reservation values for a fixed action.

Fix an action $i \in \mathcal{A}$ and define

$$g_i(z) := \mathbb{E}_{Y \sim p_i}\big[(w(Y) - z)^+\big], \qquad z \in \mathbb{R}.$$

Because $\mathcal{Y}$ is finite, g_i is continuous and nonincreasing on $\mathbb{R}$ and strictly decreasing on $(-\infty, \bar{w}_i)$, where $\bar{w}_i := \max\{w(y) : p_i(y) > 0\}$ in z. Also, $\lim_{z \to +\infty} g_i(z) = 0$, while $\lim_{z \to -\infty} g_i(z) = +\infty$. Since $c_i > 0$, any solution to $g_i(z) = c_i$ must satisfy $z < \bar{w}_i$, where strict decrease applies. Therefore, for each $c_i > 0$ there exists a unique $z_\infty(i) \in \mathbb{R}$ such that

$$g_i(z_\infty(i)) = \mathbb{E}_{Y \sim p_i}\big[(w(Y) - z_\infty(i))^+\big] = c_i. \tag{B-3}$$

Let $\bar{z}_\infty := \max_{i \in \mathcal{A}} z_\infty(i)$ and choose $i^* \in \arg\max_{i \in \mathcal{A}} z_\infty(i)$, breaking ties in the principal's favor.

Step 3: Closed-form solution of (B-2).

Define a candidate value function

$$\widetilde{V}(b) := \max\{b, \bar{z}_\infty\}, \qquad b \in \mathbb{R}_+.$$

We verify that $\widetilde{V}$ satisfies the Bellman equation (B-2).

Case 1: $b \geq \bar{z}_\infty$. Then $\widetilde{V}(b) = b$ and for any action i,

$$\widetilde{V}(\max\{b, w(Y)\}) = \max\{b, w(Y)\} = b + (w(Y) - b)^+.$$

Hence, the continuation value from sampling i equals

$$-c_i + \mathbb{E}[\widetilde{V}(\max\{b, w(Y)\})] = -c_i + b + \mathbb{E}[(w(Y) - b)^+].$$

Since $b \geq \bar{z}_\infty \geq z_\infty(i)$ and $g_i(\cdot)$ is nonincreasing, $\mathbb{E}[(w(Y) - b)^+] = g_i(b) \leq g_i(z_\infty(i)) = c_i$, so the continuation value is at most b. Thus the right-hand side of (B-2) equals $b = \widetilde{V}(b)$.

Case 2: $b < \bar{z}_\infty$. Then $\widetilde{V}(b) = \bar{z}_\infty$ and for any action i,

$$\widetilde{V}(\max\{b, w(Y)\}) = \max\{\bar{z}_\infty, w(Y)\} = \bar{z}_\infty + (w(Y) - \bar{z}_\infty)^+.$$

Therefore the continuation value from sampling i is

$$-c_i + \mathbb{E}[\widetilde{V}(\max\{b, w(Y)\})] = \bar{z}_\infty - c_i + \mathbb{E}[(w(Y) - \bar{z}_\infty)^+] = \bar{z}_\infty - c_i + g_i(\bar{z}_\infty).$$

Because $\bar{z}_\infty \geq z_\infty(i)$ and g_i is nonincreasing, $g_i(\bar{z}_\infty) \leq g_i(z_\infty(i)) = c_i$, so this continuation value is at most $\bar{z}_\infty$ for every i. For $i = i^*$, we have $\bar{z}_\infty = z_\infty(i^*)$, so $g_{i^*}(\bar{z}_\infty) = c_{i^*}$ by (B-3), implying the continuation value equals $\bar{z}_\infty$. Hence the maximization over i in (B-2) yields $\bar{z}_\infty$, and thus the right-hand side equals $\max\{b, \bar{z}_\infty\} = \widetilde{V}(b)$.

Combining the two cases, $\widetilde{V}$ satisfies (B-2). We now verify that it is the value function. Since $\widetilde{V}$ satisfies (B-2), for every state b and every action i,

$$-c_i + \mathbb{E}\Big[\widetilde{V}(\max\{b, w(Y)\})\Big] \leq \widetilde{V}(b),$$

and stopping gives payoff $b \leq \widetilde{V}(b)$. Applying these two bounds recursively shows that no admissible policy can deliver expected payoff above $\widetilde{V}(b)$. Conversely, the equality cases above identify a policy that attains the bound: stop whenever $b \geq \bar{z}_\infty$, and whenever $b < \bar{z}_\infty$, sample an action $i^* \in \operatorname{argmax}_{i \in \mathcal{A}} z_\infty(i)$. Hence

$$V_\infty(b) = \widetilde{V}(b) = \max\{b, \bar{z}_\infty\} \qquad \text{for all } b \in \mathbb{R}_+.$$

Step 4: Optimal policy and the stopping threshold.

The characterization $V_\infty(b) = \max\{b, \bar{z}_\infty\}$ implies: if $b > \bar{z}_\infty$, stopping is strictly optimal; if $b < \bar{z}_\infty$, continuing is optimal, and any action in $\arg\max_i z_\infty(i)$ is optimal (ties broken in the principal's favor select i^*). At $b = \bar{z}_\infty$, stopping and sampling an action attaining $\bar{z}_\infty$ are both optimal.

There are three initial-state cases.

(1) If $\bar{z}_\infty < 0$, then the initial state $b = 0$ satisfies $b > \bar{z}_\infty$, so stopping immediately is optimal.

(2) If $\bar{z}_\infty = 0$, then $b = 0$ is exactly the boundary state. Stopping and sampling i^* are both agent-optimal, so the selected best response is determined by principal-favor tie-breaking.

(3) If $\bar{z}_\infty > 0$, then $b = 0 < \bar{z}_\infty$, so sampling i^* is optimal. Thereafter, the agent samples below the cutoff and stops above it; if the current best payment equals the cutoff, stopping and continuing with i^* are both agent-optimal, and principal-favor tie-breaking selects among them.

Proof of (ii)(b).

Finally,

$$u_A(w, \sigma^*(w)) = V_\infty(0) = \max\{0, \bar{z}_\infty\}.$$

This completes the proof. □

B.2 Proof of Section 5

Proof of Lemma 2. Fix the debt contract $w = w^{\mathsf{debt}}(\cdot; \bar{\zeta})$ and write $V_k(\cdot) := V_k^w(\cdot)$ for the agent's continuation value defined in (2).

Step 1: A simplification of the continuation values.

We claim that

$$V_k(b) = b \qquad \forall\, k \in \{0, 1, \ldots, K\},\ \forall\, b \geq 0, \tag{B-4}$$

where b is the highest realized payment so far. The boundary condition (1) gives $V_0(b) = b$. Assume $V_{k-1}(b) = b$ for all $b \geq 0$ and consider any state (k, b) with $k \geq 1$. For any action $i \in \mathcal{A}$,

$$\begin{aligned} -c_i + \mathbb{E}_{Y\sim p_i}\big[V_{k-1}(\max\{b, w(Y)\})\big] &= -c_i + \mathbb{E}_{Y\sim p_i}\big[\max\{b, w(Y)\}\big] \\ &= b - c_i + \mathbb{E}_{Y\sim p_i}\big[(w(Y) - b)^+\big] \\ &\leq b - c_i + \mathbb{E}_{Y\sim p_i}\big[w(Y)\big] \\ &= b - c_i + \mathbb{E}_{Y\sim p_i}\big[(v(Y) - \bar{\zeta})^+\big], \end{aligned}$$

where the last equality is by the definition of the $\bar{\zeta}$-debt contract. Since ζ_i solves (7) and $\zeta \mapsto \mathbb{E}_{Y\sim p_i}[(v(Y) - \zeta)^+]$ is decreasing, $\bar{\zeta} \geq \zeta_i$ implies

$$\mathbb{E}_{Y\sim p_i}\big[(v(Y) - \bar{\zeta})^+\big] \leq \mathbb{E}_{Y\sim p_i}\big[(v(Y) - \zeta_i)^+\big] = c_i.$$

Therefore, for every i,

$$-c_i + \mathbb{E}_{Y\sim p_i}\big[V_{k-1}(\max\{b, w(Y)\})\big] \leq b,$$

and plugging into (2) yields $V_k(b) = \max\{b,\ \max_i(\cdot)\} = b$. This completes the induction and proves (B-4).

Step 2: Translating the payment state into the value state.

Let $\bar{v}$ be the highest realized value available. Under $w^{\mathsf{debt}}(\cdot;\bar{\zeta})$, the current best payment is $b = (\bar{v} - \bar{\zeta})^+$. If $\bar{v} > \bar{\zeta}$, then $b > 0$, and (B-4) gives $V_k(b) = b$. Since stopping yields exactly b, stopping attains the agent's value and is optimal. If $\bar{v} = \bar{\zeta}$, then $b = 0$ and the agent is indifferent between stopping and continuing, but stopping gives the principal payoff $\bar{\zeta}$ and continuation cannot improve on this upper bound under the debt contract. We therefore select stopping at $\bar{v} = \bar{\zeta}$ under the tie-breaking convention.

Step 3: Behavior below the face value.

Now fix a state with $\bar{v} < \bar{\zeta}$. The current best payment is $b = 0$, so stopping yields agent payoff 0. Continuing with action i gives

$$-c_i + \mathbb{E}_{Y\sim p_i}\big[V_{k-1}(w(Y))\big] = -c_i + \mathbb{E}_{Y\sim p_i}\big[w(Y)\big] = \mathbb{E}_{Y\sim p_i}\big[(v(Y) - \bar{\zeta})^+\big] - c_i \le 0,$$

with equality if and only if $\zeta_i = \bar{\zeta}$. Thus the agent is weakly best off choosing any $i \in \arg\max_j \zeta_j$; among these agent-optimal actions, the principal-favor tie-breaking convention selects $\bar{i}$.

After sampling $\bar{i}$, there are three cases.

- If $v(Y) > \bar{\zeta}$, then the new best payment is positive, so the agent stops by Step 2.
- If $v(Y) = \bar{\zeta}$, then the payment remains zero and the agent is indifferent. Stopping gives the principal payoff $\bar{\zeta}$, and continuation cannot yield more than $\bar{\zeta}$ under the debt contract; hence we select stopping.
- If $v(Y) < \bar{\zeta}$, then the highest realized value remains below $\bar{\zeta}$ and the current best payment remains zero. The agent is again indifferent between stopping and continuing with $\bar{i}$; among the agent-optimal continuations, the principal favors continuing whenever $k - 1 \ge 1$.

This establishes the stated selected policy. □

Proof of Theorem 1. We proceed in steps.

Step 1: Payoff representation under $w^{\mathsf{debt}}(\cdot;\bar{\zeta})$.

By Lemma 2, starting from $b = 0$ the agent repeatedly samples action $\bar{i}$ until either she sees a draw with $v(Y) \ge \bar{\zeta}$ or the cap K is exhausted. Let $Y^{(1)}, \dots, Y^{(K)}$ be i.i.d. draws from $p_{\bar{i}}$ and define

$$M_{\bar{i},K} := \max\{v(Y^{(1)}), \dots, v(Y^{(K)})\}.$$

Under the debt contract, the principal's realized payoff equals

$$v(\hat{Y}) - w^{\mathsf{debt}}(\hat{Y};\bar{\zeta}) = \min\{v(\hat{Y}), \bar{\zeta}\}.$$

On the event $\{M_{\bar{i},K} \geq \bar{\zeta}\}$, the principal's payoff equals $\bar{\zeta}$ regardless of which draw first crosses $\bar{\zeta}$, while on the event $\{M_{\bar{i},K} < \bar{\zeta}\}$ the agent exhausts the cap and $\hat{Y}$ attains the maximum value among the K draws (recall and monotonicity of $w^{\mathsf{debt}}(\cdot;\bar{\zeta})$ in v). Therefore, along every sample path the principal's payoff equals $\min\{M_{\bar{i},K}, \bar{\zeta}\}$, and hence

$$U_P^{\mathsf{debt}}(K;\bar{\zeta}) = \mathbb{E}\big[\min\{M_{\bar{i},K}, \bar{\zeta}\}\big]. \tag{B-5}$$

Step 2: Lower bound via the acceptance probability.

Since $\min\{M_{\bar{i},K}, \bar{\zeta}\} \geq \bar{\zeta} \cdot \mathbf{1}\{M_{\bar{i},K} \geq \bar{\zeta}\}$,

$$U_P^{\mathsf{debt}}(K;\bar{\zeta}) \geq \bar{\zeta}\ \Pr\{M_{\bar{i},K} \geq \bar{\zeta}\}.$$

By independence and the definition of $\bar{q}$ in (9),

$$\Pr\{M_{\bar{i},K} \geq \bar{\zeta}\} = 1 - \Pr\{v(Y^{(1)}) < \bar{\zeta}, \ldots, v(Y^{(K)}) < \bar{\zeta}\} = 1 - (1-\bar{q})^K.$$

Therefore,

$$U_P^{\mathsf{debt}}(K;\bar{\zeta}) \geq \bar{\zeta}(1 - (1-\bar{q})^K). \tag{B-6}$$

Step 3: Global upper bound on $\mathsf{SB}_P(K)$.

Fix any contract w and let $\sigma^*(w)$ be the selected best response. Write $\hat{Y}^* := \hat{Y}(w, \sigma^*(w))$ for the reported outcome induced by this selected best response, and let τ and $(a_k)_{k=1}^{\tau}$ denote the corresponding stopping time and sampled actions. Since the agent can always choose outside option directly and obtain utility 0, optimality implies $u_A(w, \sigma^*(w)) \geq 0$, i.e.,

$$\mathbb{E}[w(\hat{Y}^*)]\ \geq\ \mathbb{E}\Big[\sum_{k=1}^{\tau} c_{a_k}\Big].$$

Hence,

$$u_P(w, \sigma^*(w)) = \mathbb{E}[v(\hat{Y}^*)] - \mathbb{E}[w(\hat{Y}^*)] \leq \mathbb{E}[v(\hat{Y}^*)] - \mathbb{E}\Big[\sum_{k=1}^{\tau} c_{a_k}\Big].$$

The right-hand side is bounded above by the principal's first-best self-search value with cap K: the principal directly chooses the same adaptive sampling policy, keeps the best realized outcome, and pays the same sampling costs. This first-best value is weakly increasing in K, because an additional trial can always be left unused, and is therefore bounded above by the infinite-horizon first-best value. The infinite-horizon self-search problem is the stationary Weitzman search problem with reward $v(Y)$ and sampling costs c_i. Applying Proposition 2(ii).(b) with $w = v$, this value is $\max\{0, \max_i \zeta_i\}$. Hence

$$\mathsf{SB}_P(K) \leq \max\{0, \max_i \zeta_i\} = \bar{\zeta}, \tag{B-7}$$

where the last equality holds by the maintained assumption $\bar{\zeta} > 0$.

Step 4: Performance guarantee.

Combining (B-6) with $\mathsf{SB}_P(K) \le \bar{\zeta}$ yields

$$\frac{U_P^{\mathsf{debt}}(K;\bar{\zeta})}{\mathsf{SB}_P(K)} \ge \frac{\bar{\zeta}(1-(1-\bar{q})^K)}{\bar{\zeta}} = 1-(1-\bar{q})^K.$$

which proves the claim. Moreover, based on the argument in Step 3, we know that

$$\mathsf{FB}_P(\infty) = \mathsf{SB}_P(\infty) = U_P^{\mathsf{debt}}(\infty;\bar{\zeta}) = \bar{\zeta}.$$

Agent's payoff

By (B-4), under the $\bar{\zeta}$-debt contract, we have $V_k(b) = b$. Thus,

$$U_A^{\mathsf{debt}}(K;\bar{\zeta}) = V_K^w(0) = 0.$$

This completes the proof. □

Proof of Remark 2. Fix an integer $K \ge 2$ and consider the example specified in the remark: $\mathcal{A} = \mathcal{Y} = \{1,\dots,K\}$, $v(y) = 2^y$, $p_{ii} = 1$, and $c_i = 2^i - i$.

Step 1: Computing $\bar{\zeta}$ and the $\bar{\zeta}$-debt payoff.

For each action i, $v(Y) = 2^i$ almost surely, so the Weitzman index ζ_i defined by

$$\mathbb{E}_{Y\sim p_i}\left[(v(Y)-\zeta_i)^+\right] = c_i$$

satisfies $(2^i-\zeta_i)^+ = 2^i - i$. Since $c_i > 0$, we must have $\zeta_i < 2^i$, and hence the equation becomes $2^i - \zeta_i = 2^i - i$, yielding the unique solution $\zeta_i = i$. Therefore

$$\bar{\zeta} = \max_{i\in\mathcal{A}} \zeta_i = K.$$

Now consider the $\bar{\zeta}$-debt contract $w^{\mathsf{debt}}(\cdot;\bar{\zeta}) = w^{\mathsf{debt}}(\cdot;K)$, i.e., $w^{\mathsf{debt}}(y;K) = (2^y - K)^+$. If the agent takes action K once, then she receives payment $w^{\mathsf{debt}}(K;K) = 2^K - K$ and pays cost $c_K = 2^K - K$, so her utility is 0, equal to the outside option from choosing $\tau = 0$; by tie-breaking in the principal's favor she takes the action. Since the realized payment is strictly positive, by Lemma 2 she stops immediately and reports $Y = K$. Thus the principal's realized payoff is

$$v(K) - w^{\mathsf{debt}}(K;K) = 2^K - (2^K - K) = K,$$

so for this instance and sampling budget K,

$$U_P^{\mathsf{debt}}(K;\bar{\zeta}) = U_P^{\mathsf{debt}}(K;K) = K.$$

Step 2: Under a linear contract, the agent never needs more than one trial.

Fix $\alpha \in [0,1]$ and consider $w_\alpha^{\mathsf{lin}}(y) = \alpha v(y) = \alpha 2^y$. Let σ be any policy with stopping time $\tau \le K$ and sampled actions $(a_1,\dots,a_\tau)$. If $\tau = 0$, the agent payoff is 0. If $\tau \ge 1$, then outcomes are

deterministic and $w_\alpha^{\mathsf{lin}}(\cdot)$ is weakly increasing in y; by principal-favor tie-breaking among payment ties, recall implies that the reported outcome is the maximum sampled index

$$\hat{Y}(w_\alpha^{\mathsf{lin}}, \sigma) = \max\{a_1, \ldots, a_\tau\} =: j, \qquad \text{and} \qquad w_\alpha^{\mathsf{lin}}(j) = \alpha 2^j.$$

Moreover, since action j is sampled at least once, the total incurred cost satisfies $\sum_{k=1}^{\tau} c_{a_k} \geq c_j$. Hence

$$u_A(w_\alpha^{\mathsf{lin}}, \sigma) = \alpha 2^j - \sum_{k=1}^{\tau} c_{a_k} \;\leq\; \alpha 2^j - c_j.$$

The upper bound is attained by the one-step policy that samples action j once and then stops. Therefore, under w_α^{lin}, there is a selected best response that chooses either $\tau = 0$, or $\tau = 1$ with some action

$$i^*(\alpha) \in \arg\max_{i \in \{1,\ldots,K\}} \{\alpha 2^i - c_i\}, \tag{B-8}$$

whenever the maximum is nonnegative (and $\tau = 0$ otherwise).

Step 3: Upper bound $U_P^{\mathsf{lin}}(K;\alpha) \leq 2$ for all α.

If the agent chooses $\tau = 0$, then $U_P^{\mathsf{lin}}(K;\alpha) = 0 \leq 2$. Otherwise, letting $i = i^*(\alpha)$, the principal payoff is

$$U_P^{\mathsf{lin}}(K;\alpha) = v(i) - w_\alpha^{\mathsf{lin}}(i) = (1-\alpha)2^i.$$

If $i = 1$ then $(1-\alpha)2 \leq 2$. If $i \geq 2$, optimality of i, see (B-8), implies

$$\alpha 2^i - c_i \;\geq\; \alpha 2^{i-1} - c_{i-1}.$$

Substituting $c_\ell = 2^\ell - \ell$ yields

$$\alpha 2^i - (2^i - i) \;\geq\; \alpha 2^{i-1} - (2^{i-1} - (i-1)) \quad \Longleftrightarrow \quad 1 \;\geq\; (1-\alpha)2^{i-1}.$$

Multiplying by 2 gives $(1-\alpha)2^i \leq 2$. Hence for every $\alpha \in [0,1]$,

$$U_P^{\mathsf{lin}}(K;\alpha) \leq 2 \qquad \Longrightarrow \qquad U_{P,*}^{\mathsf{lin}}(K) = \max_{\alpha \in [0,1]} U_P^{\mathsf{lin}}(K;\alpha) \leq 2.$$

Step 4: Achievability and the optimal linear contract.

We will show that

$$\alpha^* = 1 - 2^{-(K-1)} \in (0,1).$$

If the agent chooses action K once, the principal payoff is

$$(1-\alpha^*)2^K = 2^{-(K-1)}2^K = 2.$$

We now verify that under $w_{\alpha^*}^{\mathsf{lin}}$ the agent (by tie-breaking) indeed chooses action K. The agent

utility from choosing action i once equals

$$\alpha^* 2^i - c_i = \alpha^* 2^i - (2^i - i) = i - (1-\alpha^*)2^i = i - 2^{i-(K-1)}.$$

For each $i = 2, \ldots, K$, the increment from $i-1$ to i is

$$\left(i - 2^{i-(K-1)}\right) - \left((i-1) - 2^{i-1-(K-1)}\right) = 1 - 2^{i-K} \;\geq\; 0,$$

with equality if and only if $i = K$. Thus the maximizers are $\{K-1, K\}$. The principal's payoff is 2 if the agent selects K and 1 if she selects $K-1$; hence principal-favor tie-breaking selects K, giving $U_P^{\mathsf{lin}}(K; \alpha^*) = 2$. Therefore $U_{P,*}^{\mathsf{lin}}(K) \geq 2$, which together with Step 3 implies

$$U_{P,*}^{\mathsf{lin}}(K) = 2.$$

Combining Step 1 and Step 4 yields, for this instance,

$$\frac{U_{P,*}^{\mathsf{lin}}(K)}{U_P^{\mathsf{debt}}(K; \bar{\zeta})} = \frac{2}{K} \;\xrightarrow[K\to\infty]{}\; 0,$$

as claimed. □

Proof of Proposition 3. The algorithm is stated in Algorithm 1, and its supporting subroutines are Algorithms 2 to 4. See Section A for the convention $\mathcal{Y}_0 := \{0\} \cup \mathcal{Y}$ with $v(0) := 0$ and the incumbent state $s \in \mathcal{Y}_0$. Let

$$v_{\max} := \max_{y \in \mathcal{Y}} v(y).$$

If $v_{\max} = 0$, every debt contract gives the principal payoff zero, and $x = 0$ is optimal. Hence suppose $v_{\max} > 0$.

Step 1: Restrict the face value to $[0, v_{\max}]$. If $x > v_{\max}$, then $(v(y) - x)^+ = 0$ for every outcome y. Since all actions have strictly positive costs, the agent strictly prefers not to sample, and the principal payoff is zero. Thus no optimal face value is lost by restricting attention to $x \in [0, v_{\max}]$.

Step 2: Bellman recursion. Fix $x \in [0, v_{\max}]$. Under the debt contract $w^{\mathsf{debt}}(y; x) = (v(y) - x)^+$, payments are weakly increasing in $v(y)$. With recall, the agent reports an outcome with maximal payment; under principal-favor tie-breaking, among payment-maximizing reports the principal receives the highest value. Hence the payoff-relevant history is summarized by the incumbent outcome $s \in \mathcal{Y}_0$ and the number k of remaining trials.

For each state (k, s), let $(A_k(s; x), P_k(s; x))$ be the continuation payoff pair selected by maximizing the agent payoff and, among agent-optimal continuations, maximizing the principal payoff. For $k = 0$,

$$A_0(s; x) = (v(s) - x)^+, \qquad P_0(s; x) = \min\{v(s), x\}.$$

For $k \geq 1$, the agent either stops immediately or chooses an action i. If action i is chosen, the

payoff pair is

$$\left(-c_i + \sum_{y\in\mathcal{Y}} p_i(y)\, A_{k-1}(\max\{s,y\};x),\ \sum_{y\in\mathcal{Y}} p_i(y)\, P_{k-1}(\max\{s,y\};x)\right).$$

Taking the lexicographic maximum[5] over the stop option and all action candidates gives the equilibrium continuation pair at (k,s). The game starts from $(K,0)$, so $U_P^{\mathsf{debt}}(K;x) = P_K(0;x)$.

Step 3: Breakpoints. We show by induction on k that, for every $s \in \mathcal{Y}_0$, both $A_k(s;\cdot)$ and $P_k(s;\cdot)$ are piecewise affine on $[0, v_{\max}]$ with finite breakpoint sets. View the face value x as the variable. At $k=0$, both $(v(s)-x)^+$ and $\min\{v(s),x\}$ are affine between the breakpoints $\{0, v(s), v_{\max}\}$. Suppose the value functions at layer $k-1$ have finite breakpoint representations. For any action i, the action value at layer k is a weighted sum of next-state values, so it is affine on the common refinement of the next-state breakpoint lists. The stop option is also piecewise affine. The agent's best response is obtained by taking the pointwise lexicographic envelope of these finitely many candidates. Two affine candidates can change their lexicographic order only at finitely many crossing points. If their agent-payoff affine pieces coincide on an interval, then they are agent-tied throughout that interval and the lexicographic comparison reduces to their principal-payoff affine pieces; if those also coincide, the two candidates are equivalent for tie-breaking on that interval. Thus adding the relevant agent-payoff and principal-payoff crossings gives a finite breakpoint representation at layer k. By induction, $P_K(0;\cdot)$ is piecewise affine on $[0, v_{\max}]$.

Moreover, it suffices to search over the final breakpoint list. On each open interval between consecutive breakpoints, the selected candidate is fixed and $P_K(0;x)$ is affine, so the supremum over that interval is approached at an endpoint. Each candidate payoff pair is continuous in x, and at every breakpoint the algorithm stores the pointwise lexicographic maximum. Thus the principal payoff at the endpoint is at least the corresponding one-sided limit, and no interior point is needed.

Step 4: Running time. Let $N_k(s)$ be the number of open affine intervals in the representation of $(A_k(s;\cdot), P_k(s;\cdot))$, and define

$$L_k := \max_{s\in\mathcal{Y}_0} N_k(s).$$

That is, L_k is a worst-case *per-cell* representation size, where a cell means one state pair (k,s). This is the relevant quantity because Algorithm 1 processes one cell (k,s) at a time; the total number of cells, $K(m+1)$, is multiplied in only after bounding the work required for one cell.

For each fixed s, the $k=0$ representation is produced by Algorithm 4: it represents the two functions $(v(s)-x)^+$ and $\min\{v(s),x\}$, which have at most one kink, at $x=v(s)$. Hence $N_0(s) \le 2$ for every s, and taking the maximum over $s \in \mathcal{Y}_0$ gives $L_0 \le 2$.

Fix a cell (k,s) with $k \ge 1$. For one action i, Algorithm 2 looks up the m next-state represen-

[5] Taking the lexicographic maximum means first maximizing the agent payoff, the first component of the pair, and then, among agent-payoff ties, choosing the alternative with the largest principal payoff, the second component.

tations $\mathcal{R}_{k-1}(\max\{s, y\})$, one for each $y \in \mathcal{Y}$. Their common refinement has at most mL_{k-1} open intervals, so the action-i candidate is affine on at most mL_{k-1} intervals. On each interval in this refinement, the relevant next-state formula is affine for every y, so the two affine coefficients of the action-i candidate are obtained by taking the weighted sums in Algorithm 2. Hence building the action-i candidate requires one pass through the refinement, and its cost is linear in the number of intervals.

At cell (k, s), Algorithm 1 then passes to Algorithm 3 a list of $n + 1$ candidates: one stop candidate and one action candidate for each of the n actions. The stop candidate contributes at most two intervals, while the n action candidates contribute at most nmL_{k-1} intervals before pairwise crossings are inserted. Hence the initial common refinement entering the envelope routine has at most $2+nmL_{k-1}$ open intervals. On each such interval, every candidate pair $(A^r(x), P^r(x))$ is affine. Algorithm 3 checks all pairs $1 \le r < \ell \le n + 1$; there are $O((n + 1)^2)$ such pairs, and each pair can add only a constant number of crossing points on a fixed interval. Therefore, for a universal constant C,

$$L_k \le C(n + 1)^2(2 + nmL_{k-1}).$$

Since $n, m \ge 1$ and $L_{k-1} \ge 1$, the recurrence implies

$$L_k = O(n^3 mL_{k-1}).$$

Starting from $L_0 = O(1)$, iteration gives

$$L_k = O((n^3 m)^k) = O(n^{3k} m^k).$$

Thus

$$L_{K-1} = O(n^{3K-3} m^{K-1}), \qquad L_K = O(n^{3K} m^K),$$

for fixed K.

The recurrence above bounds the number of breakpoints *stored* at each cell. The *time* to process a cell is larger than this output size by one additional factor of order n, because the winner-selection step in Algorithm 3 compares all $n + 1$ candidates on each crossing subinterval. More explicitly, within each of the $O(2 + nmL_{k-1})$ common-refinement intervals, the routine creates $O((n + 1)^2)$ crossing subintervals, and on each such subinterval it chooses the lexicographic winner by comparing all $n + 1$ candidates. Hence processing one cell takes $O((n + 1)^3(2 + nmL_{k-1}))$ time. Processing all states and layers takes

$$O\Big(K(m + 1)(n + 1)^3(2 + nmL_{K-1})\Big) = O(n^{3K+1} m^{K+1})$$

for fixed K. The final scan over the breakpoint list $B_K(0)$ costs $O(L_K) = O(n^{3K} m^K)$, so it is dominated by the dynamic-programming construction above. This proves the claimed running

time and completes the proof. □

Proof of Proposition 4. Fix $K \geq 1$ and for notation simplicity write $x^* := x^*_{\mathsf{debt}}$.

Step 1: $x^* \leq \bar{\zeta}$.

Fix any $x > \bar{\zeta}$ and consider the debt contract $w^{\mathsf{debt}}(\cdot; x)$. Since $\zeta_i \leq \bar{\zeta} < x$ and $\zeta \mapsto \mathbb{E}_{Y\sim p_i}[(v(Y) - \zeta)^+]$ is decreasing,

$$\mathbb{E}_{Y\sim p_i}[w^{\mathsf{debt}}(Y; x)] = \mathbb{E}_{Y\sim p_i}[(v(Y) - x)^+] < \mathbb{E}_{Y\sim p_i}[(v(Y) - \zeta_i)^+] = c_i \qquad \forall\, i \in \mathcal{A}.$$

The strict inequality follows because the defining equation for ζ_i makes the expectation at ζ_i equal to the positive cost c_i, so there is positive probability of outcomes above ζ_i; raising the threshold from ζ_i to $x > \zeta_i$ strictly reduces the expectation. Let σ be any agent policy under $w^{\mathsf{debt}}(\cdot; x)$, with stopping time $\tau \leq K$ and sampled action sequence $(a_1, \ldots, a_\tau)$ on histories with $\tau \geq 1$. For sampled trial k, write $\pi_k := (v(Y_k) - x)^+$, and define the recalled payment by

$$\Pi_\tau := \begin{cases} 0, & \tau = 0, \\ \max_{1\leq s\leq \tau} \pi_s, & \tau \geq 1. \end{cases}$$

Then $\Pi_\tau \leq \sum_{s=1}^{\tau} \pi_s$ pathwise, so the agent's expected utility satisfies

$$u_A(w^{\mathsf{debt}}(\cdot; x), \sigma) = \mathbb{E}\Big[\Pi_\tau - \sum_{s=1}^{\tau} c_{a_s}\Big] \leq \mathbb{E}\Big[\sum_{s=1}^{\tau} \pi_s - \sum_{s=1}^{\tau} c_{a_s}\Big].$$

We have

$$\mathbb{E}\Big[\sum_{s=1}^{\tau}(\pi_s - c_{a_s})\Big] = \sum_{s=1}^{K} \mathbb{E}[\mathbf{1}\{s \leq \tau\}(\pi_s - c_{a_s})].$$

At trial s, the event $\{s \leq \tau\}$ and the action a_s are determined before the draw Y_s is observed. Hence, by iterated expectations and the strict bound above,

$$\mathbb{E}[\mathbf{1}\{s \leq \tau\}(\pi_s - c_{a_s})] = \mathbb{E}\Big[\mathbf{1}\{s \leq \tau\}\Big(\mathbb{E}_{Y\sim p_{a_s}}[(v(Y) - x)^+] - c_{a_s}\Big)\Big] \leq 0.$$

Summing over $s = 1, \ldots, K$ yields

$$\mathbb{E}\Big[\sum_{s=1}^{\tau} \pi_s - \sum_{s=1}^{\tau} c_{a_s}\Big] \leq 0.$$

Moreover, the inequality is strict for any policy that samples with positive probability, i.e., any policy with $\Pr(\tau \geq 1) > 0$. Hence the unique optimal response is to stop immediately with $\tau = 0$, and the principal payoff is $U_P^{\mathsf{debt}}(K; x) = 0$. Since $U_P^{\mathsf{debt}}(K; \bar{\zeta}) > 0$ whenever $\bar{\zeta} > 0$, a face value $x > \bar{\zeta}$ cannot be optimal, and hence $x^* \leq \bar{\zeta}$.

Step 2: A lower bound on x^*.

For any $x \geq 0$, the principal's realized payoff under $w^{\mathsf{debt}}(\cdot; x)$ equals $\min\{v(\hat{Y}), x\} \leq x$, hence

$$U_P^{\mathsf{debt}}(K; x) \leq x. \tag{B-9}$$

By optimality of x^* and (B-9),

$$x^* \;\geq\; U_P^{\mathsf{debt}}(K; x^*) \;\geq\; U_P^{\mathsf{debt}}(K; \bar{\zeta}) \;\geq\; \bar{\zeta}(1 - (1 - \bar{q})^K),$$

where the last inequality follows from (B-6).

Step 3: Concluding the face-value distance bound.

Combining Step 1 and Step 2 gives

$$0 \leq \bar{\zeta} - x^* \leq \bar{\zeta} - \bar{\zeta}(1 - (1 - \bar{q})^K) = \bar{\zeta}(1 - \bar{q})^K,$$

which proves the first claim.

Step 4: Bounding the payoff gap $U_{P,*}^{\mathsf{debt}}(K) - U_P^{\mathsf{debt}}(K; \bar{\zeta})$.

The lower bound $0 \leq U_{P,*}^{\mathsf{debt}}(K) - U_P^{\mathsf{debt}}(K; \bar{\zeta})$ holds by optimality of x^*. For the upper bound, first note that

$$U_{P,*}^{\mathsf{debt}}(K) = U_P^{\mathsf{debt}}(K; x^*) \leq x^* \leq \bar{\zeta},$$

where the first inequality is (B-9) and the second is Step 1. Combining this global upper bound with the lower bound (B-6) for the $\bar{\zeta}$-debt contract gives

$$U_{P,*}^{\mathsf{debt}}(K) - U_P^{\mathsf{debt}}(K; \bar{\zeta}) \;\leq\; \bar{\zeta} - \bar{\zeta}(1 - (1 - \bar{q})^K) = \bar{\zeta}(1 - \bar{q})^K,$$

which proves the second claim. □

Proof of Proposition 5. Fix $x \geq 0$ and write

$$w_x(y) := w^{\mathsf{debt}}(y; x) = (v(y) - x)^+.$$

For any feasible policy σ, let τ be its stopping time and define

$$M_\sigma := \begin{cases} 0, & \tau = 0, \\ \max_{1 \leq t \leq \tau} v(Y_t), & \tau \geq 1. \end{cases}$$

Because $w_x(\cdot)$ is nondecreasing in $v(\cdot)$ and recall is allowed, the reported outcome attains M_σ (if all payments are 0, the report tie is broken in the principal's favor). Hence, pathwise,

$$v(\hat{Y}) - w_x(\hat{Y}) = M_\sigma - (M_\sigma - x)^+ = M_\sigma \wedge x := \min\{M_\sigma, x\},$$

and

$$w_x(\hat{Y}) - \sum_{t=1}^{\tau} c_{a_t} = (M_\sigma - x)^+ - \sum_{t=1}^{\tau} c_{a_t}.$$

Therefore, for every feasible policy σ,

$$u_P(w_x, \sigma) = \mathbb{E}[M_\sigma \wedge x], \qquad u_A(w_x, \sigma) = \mathbb{E}\left[(M_\sigma - x)^+ - \sum_{t=1}^{\tau} c_{a_t}\right]. \tag{B-10}$$

Continuation values. For $k \in \mathbb{Z}_+$ and current running maximum value $\bar{v} \geq 0$, let $A_k(\bar{v})$ be the agent's maximal continuation payoff with k trials remaining, and let $P_k(\bar{v})$ be the principal's continuation payoff under principal-favor tie-breaking among agent-optimal continuation policies. Let

$$V_i := v(Y) \qquad \text{for } Y \sim p_i.$$

Then

$$A_0(\bar{v}) = (\bar{v} - x)^+, \qquad P_0(\bar{v}) = \bar{v} \wedge x.$$

For $k \geq 1$, define

$$A_k^{\text{stop}}(\bar{v}) = (\bar{v} - x)^+, \qquad P_k^{\text{stop}}(\bar{v}) = \bar{v} \wedge x,$$

and for each $i \in \mathcal{A}$,

$$A_k^i(\bar{v}) = -c_i + \mathbb{E}[A_{k-1}(\bar{v} \vee V_i)], \qquad P_k^i(\bar{v}) = \mathbb{E}[P_{k-1}(\bar{v} \vee V_i)].$$

Hence

$$A_k(\bar{v}) = \max\Big\{A_k^{\text{stop}}(\bar{v}),\ \max_{i \in \mathcal{A}} A_k^i(\bar{v})\Big\}.$$

Let

$$\mathcal{I}_k(\bar{v}) := \arg\max_{a \in \{\text{stop}\} \cup \mathcal{A}} A_k^a(\bar{v}).$$

Then principal-favor tie-breaking implies

$$P_k(\bar{v}) = \max\{P_k^a(\bar{v}) : a \in \mathcal{I}_k(\bar{v})\}.$$

In particular,

$$U_A^{\mathsf{debt}}(K; x) = A_K(0), \qquad U_P^{\mathsf{debt}}(K; x) = P_K(0).$$

Step 1: $A_k(\cdot)$ is constant on $[0, x]$.

We prove by induction on k that for every $k \geq 0$, $A_k(\cdot)$ is constant on $[0, x]$.

For $k = 0$, this is immediate because $A_0(\bar{v}) = (\bar{v} - x)^+ = 0$ for all $\bar{v} \in [0, x]$.

Now fix $k \geq 1$ and suppose $A_{k-1}(\cdot)$ is constant on $[0, x]$. Take any $\bar{v} \in [0, x]$ and any $i \in \mathcal{A}$. If $V_i > x$, then $\bar{v} \vee V_i = V_i$; if $V_i \leq x$, then $\bar{v} \vee V_i \in [0, x]$. Hence, $A_{k-1}(\bar{v} \vee V_i)$ does not depend on $\bar{v} \in [0, x]$. Therefore $A_k^i(\bar{v})$ is constant on $[0, x]$ for each $i \in \mathcal{A}$. Also $A_k^{\text{stop}}(\bar{v}) = 0$ on $[0, x]$. Taking the maximum over stop and $i \in \mathcal{A}$ shows that $A_k(\cdot)$ is constant on $[0, x]$.

Step 2: $P_k(\cdot)$ is nondecreasing on $\mathbb{R}_+$.

We again proceed by induction on k.

For $k = 0$, $P_0(\bar{v}) = \bar{v} \wedge x$ is nondecreasing.

Now fix $k \geq 1$ and assume $P_{k-1}(\cdot)$ is nondecreasing. For each $i \in \mathcal{A}$, the map

$$\bar{v} \mapsto P_k^i(\bar{v}) = \mathbb{E}[P_{k-1}(\bar{v} \vee V_i)]$$

is nondecreasing, because $\bar{v} \vee V_i$ is nondecreasing in $\bar{v}$ pointwise and P_{k-1} is nondecreasing. Also $P_k^{\text{stop}}(\bar{v}) = \bar{v} \wedge x$ is nondecreasing.

If $\bar{v} \geq x$, then the current running maximum is already at least x, so under any continuation policy the principal's realized payoff is

$$M_\sigma \wedge x = x$$

pathwise. Hence

$$P_k(\bar{v}) = x \qquad \text{for all } \bar{v} \geq x.$$

Now take $0 \leq \bar{v} \leq \bar{u}$. If $\bar{u} \leq x$, then by Step 1, for every $a \in \{\text{stop}\} \cup \mathcal{A}$, the quantity $A_k^a(u)$ is constant for $u \in [0, x]$. Therefore the maximizing set $\mathcal{I}_k(u)$ is the same for all $u \in [0, x]$; denote this common set by $\mathcal{I}_k^0$. Hence

$$P_k(\bar{v}) = \max_{a \in \mathcal{I}_k^0} P_k^a(\bar{v}) \leq \max_{a \in \mathcal{I}_k^0} P_k^a(\bar{u}) = P_k(\bar{u}),$$

because each $P_k^a(\cdot)$ is nondecreasing.

If instead $\bar{u} \geq x$, then $P_k(\bar{u}) = x$, while always $P_k(\bar{v}) \leq x$ because the principal's payoff is pathwise bounded above by x. Thus $P_k(\bar{v}) \leq P_k(\bar{u})$ also in this case. Therefore $P_k(\cdot)$ is nondecreasing on $\mathbb{R}_+$.

Step 3: statewise horizon monotonicity of the principal payoff.

We claim that for every $k \geq 0$ and every $\bar{v} \geq 0$,

$$P_{k+1}(\bar{v}) \geq P_k(\bar{v}). \tag{B-11}$$

Fix $k \geq 0$ and $\bar{v} \geq 0$.

For any continuation policy σ feasible from state $(k, \bar{v})$, let

$$U_A^k(\bar{v}; \sigma) \qquad \text{and} \qquad U_P^k(\bar{v}; \sigma)$$

denote the agent's and principal's continuation payoffs, respectively.

Now choose a continuation policy $\sigma^\star$ from state $(k+1, \bar{v})$ that is optimal for the agent and, among all such agent-optimal continuation policies, maximizes the principal's payoff. Thus

$$U_A^{k+1}(\bar{v}; \sigma^\star) = A_{k+1}(\bar{v}), \qquad U_P^{k+1}(\bar{v}; \sigma^\star) = P_{k+1}(\bar{v}).$$

Let a denote the *first move* prescribed by $\sigma^\star$. Hence:

$$a = \text{stop} \quad \text{means that } \sigma^\star \text{ stops immediately at state } (k+1, \bar{v}),$$

whereas $a = i \in \mathcal{A}$ means that $\sigma^\star$ first samples action i and then, after the realized next state, follows a principal-favored agent-optimal continuation policy.

Case 1: $a = i \in \mathcal{A}$.

Since $\sigma^\star$ first samples action i, the Bellman recursion gives

$$P_{k+1}(\bar{v}) = P^i_{k+1}(\bar{v}) = \mathbb{E}[P_k(\bar{v} \vee V_i)]\,.$$

Because $\bar{v} \vee V_i \geq \bar{v}$ pointwise and $P_k(\cdot)$ is nondecreasing by Step 2,

$$P_{k+1}(\bar{v}) = \mathbb{E}[P_k(\bar{v} \vee V_i)] \geq P_k(\bar{v}).$$

Case 2: $a = \text{stop}$.

In this case $\sigma^\star$ stops immediately. Therefore

$$A_{k+1}(\bar{v}) = A^{\text{stop}}_{k+1}(\bar{v}) = (\bar{v} - x)^+, \qquad P_{k+1}(\bar{v}) = P^{\text{stop}}_{k+1}(\bar{v}) = \bar{v} \wedge x.$$

We first show that

$$A_k(\bar{v}) = (\bar{v} - x)^+.$$

Indeed, every continuation policy feasible with k trials is also feasible with $k+1$ trials, so

$$A_k(\bar{v}) \leq A_{k+1}(\bar{v}) = (\bar{v} - x)^+.$$

On the other hand, stopping is feasible with k trials, so

$$A_k(\bar{v}) \geq (\bar{v} - x)^+.$$

Hence

$$A_k(\bar{v}) = (\bar{v} - x)^+.$$

Since stopping itself attains the agent value $A_k(\bar{v})$, it is agent-optimal at horizon k. Therefore

$$P_k(\bar{v}) \geq P^{\text{stop}}_k(\bar{v}) = \bar{v} \wedge x.$$

We now prove that the equality holds. Suppose, for contradiction, that

$$P_k(\bar{v}) > \bar{v} \wedge x.$$

Then, by the definition of $P_k(\bar{v})$, there exists a horizon-k continuation policy σ from state $(k, \bar{v})$ such that

$$U^k_A(\bar{v}; \sigma) = A_k(\bar{v}), \qquad U^k_P(\bar{v}; \sigma) = P_k(\bar{v}).$$

So σ is agent-optimal at state $(k, \bar{v})$, and among all such agent-optimal policies it gives the principal payoff $P_k(\bar{v})$.

Now consider the horizon-$(k+1)$ policy $\tilde{\sigma}$ that simply follows σ exactly and never uses the extra available trial. Because $\tilde{\sigma}$ generates exactly the same actions, stopping time, realized outcomes, and costs as σ, it yields the same payoffs:

$$U_A^{k+1}(\bar{v}; \tilde{\sigma}) = U_A^k(\bar{v}; \sigma) = A_k(\bar{v}) = (\bar{v} - x)^+ = A_{k+1}(\bar{v}),$$

and

$$U_P^{k+1}(\bar{v}; \tilde{\sigma}) = U_P^k(\bar{v}; \sigma) = P_k(\bar{v}).$$

The first display shows that $\tilde{\sigma}$ gives the agent exactly the maximal horizon-$(k+1)$ payoff $A_{k+1}(\bar{v})$. Hence $\tilde{\sigma}$ is also agent-optimal at state $(k+1, \bar{v})$.

But its principal payoff is

$$U_P^{k+1}(\bar{v}; \tilde{\sigma}) = P_k(\bar{v}) > \bar{v} \wedge x = P_{k+1}^{\text{stop}}(\bar{v}).$$

Thus, among agent-optimal continuation policies at state $(k+1, \bar{v})$, the policy $\tilde{\sigma}$ gives the principal strictly more than stopping. This contradicts the choice of $\sigma^\star$, which was defined to maximize the principal's payoff among all agent-optimal horizon-$(k+1)$ continuation policies.

Therefore $P_k(\bar{v}) \leq \bar{v} \wedge x$. Combining this with the earlier lower bound $P_k(\bar{v}) \geq \bar{v} \wedge x$, we conclude that

$$P_k(\bar{v}) = \bar{v} \wedge x = P_{k+1}(\bar{v}).$$

In either case, $P_{k+1}(\bar{v}) \geq P_k(\bar{v})$, proving (B-11).

Setting $\bar{v} = 0$ yields

$$U_P^{\mathsf{debt}}(K+1; x) = P_{K+1}(0) \geq P_K(0) = U_P^{\mathsf{debt}}(K; x),$$

so the principal's payoff is weakly increasing in K for every fixed debt face value x.

Step 4: monotonicity of the agent payoff and of the optimal-debt payoff.

For every $k \geq 0$ and every $\bar{v} \geq 0$, every k-trial policy is also feasible with $k+1$ trials available, so

$$A_{k+1}(\bar{v}) \geq A_k(\bar{v}).$$

Setting $\bar{v} = 0$ yields

$$U_A^{\mathsf{debt}}(K+1; x) = A_{K+1}(0) \geq A_K(0) = U_A^{\mathsf{debt}}(K; x).$$

Finally, define

$$U_{P,*}^{\mathsf{debt}}(K) := \sup_{x \geq 0} U_P^{\mathsf{debt}}(K; x).$$

Since $U_P^{\mathsf{debt}}(K; x)$ is nondecreasing in K for each fixed x, taking the supremum over $x \geq 0$ preserves the inequality:

$$U_{P,*}^{\mathsf{debt}}(K+1) \geq U_{P,*}^{\mathsf{debt}}(K).$$

This proves all monotonicity statements in Proposition 5. The final claim that $U_A^{\mathsf{debt}}(K; x_{\mathsf{debt}}^*(K))$ need not be monotone in K is a separate counterexample in Figure 1. □

B.3 Proof of Section 6

B.3.1 Proof of Section 6.1

Proof of Proposition 6. Fix $K \geq 1$, and let $W^K = (w_t^K)_{t=1}^K$ be an optimal time-contingent contract under cap K. Define a contract for cap $K+1$ by extending W^K as follows:

$$\widetilde{W}^{K+1} := (\widetilde{w}_t)_{t=1}^{K+1}, \qquad \widetilde{w}_t := w_t^K \quad (t \leq K), \qquad \widetilde{w}_{K+1}(y) := 0 \quad \forall y \in \mathcal{Y}.$$

Step 1: The agent never stops at $K+1$ under $\widetilde{W}^{K+1}$. At any history after K trials, using the extra trial yields zero final payment and incurs a positive cost, while stopping immediately yields a nonnegative payment and avoids that cost. Hence no best response uses the added trial, so $\sigma_{K+1}^*(\widetilde{W}^{K+1})$ satisfies $\tau \leq K$ almost surely.

Step 2: The induced equilibrium under $\widetilde{W}^{K+1}$ coincides with the cap-K equilibrium under W^K. By Step 1, only stopping times $\tau \in \{0, 1, \ldots, K\}$ occur under $\sigma_{K+1}^*(\widetilde{W}^{K+1})$. On this event, the realized payments are governed by $\widetilde{w}_t = w_t^K$ for $t \leq K$. Thus the agent and principal payoffs from every relevant policy are the same under $\widetilde{W}^{K+1}$ and under W^K, and the principal-favor tie-breaking rule gives the same principal payoff. Consequently,

$$u_P(\widetilde{W}^{K+1}, \sigma_{K+1}^*(\widetilde{W}^{K+1})) = u_P(W^K, \sigma_K^*(W^K)) = \mathsf{SB}_P^\tau(K).$$

Step 3: Conclude monotonicity. Since $\widetilde{W}^{K+1}$ is feasible under cap $K+1$, we have

$$\mathsf{SB}_P^\tau(K+1) \;\geq\; u_P(\widetilde{W}^{K+1}, \sigma_{K+1}^*(\widetilde{W}^{K+1})) = \mathsf{SB}_P^\tau(K).$$

□

Proof of Corollary 1. This directly follows from Theorem 1, where the performance bound holds against the first-best benchmark. □

B.3.2 Proof of Section 6.2

We use the following auxiliary lemma to facilitate the proof of Theorem 2.

Lemma B-1 (State reduction under nondecreasing contracts)**.** *Fix a contract $w : \mathcal{Y} \to \mathbb{R}_+$ that is nondecreasing in the outcome. At any history with incumbent s, the agent's optimal stopping payoff is $w(s)$. Consequently, the agent's continuation value $A_w(s)$ depends only on s and is the* unique *bounded solution to*

$$A_w(s) = \max\Big\{w(s),\ \max_{i\in\mathcal{A}}\Big[-c_i + \sum_{z\in\mathcal{Y}} P_i(z \mid s)\, A_w(s\vee z)\Big]\Big\}. \tag{B-12}$$

Proof of Theorem 2. Write $F^\star := J(0)$ throughout.

Step 1: State reduction and the agent's Bellman equation under $w^\star$.

Since v is strictly increasing, the debt contract $w^\star(y) := w^{\mathsf{debt}}(y; J(0)) = (v(y) - F^\star)^+$ is nondecreasing in y. Applying Lemma B-1 to $w^\star$, the agent's continuation value $A^\star : \mathcal{Y} \to \mathbb{R}_+$ is the unique bounded solution to

$$A^\star(s) = \max\Big\{(v(s) - F^\star)^+,\ \max_{i\in\mathcal{A}}\Big[-c_i + \sum_{z\in\mathcal{Y}} P_i(z \mid s)\, A^\star(s\vee z)\Big]\Big\}, \qquad s \in \mathcal{Y}. \tag{B-13}$$

Step 2: Identification of the agent's value function.

Define

$$B(s) := J(s) - F^\star, \qquad s \in \mathcal{Y}.$$

By the assumption of the theorem that

$$J(s) \geq J(0) = F^\star \qquad \forall s \in \mathcal{Y},$$

we have

$$B(s) \geq 0 \qquad \forall s \in \mathcal{Y}. \tag{B-14}$$

Now start from the first-best Bellman equation (12):

$$J(s) = \max\Big\{v(s),\ \max_{i\in\mathcal{A}}\Big[-c_i + \sum_{z\in\mathcal{Y}} P_i(z \mid s)\, J(s\vee z)\Big]\Big\}.$$

Subtracting the constant $F^\star$ from both sides yields

$$\begin{aligned} B(s) &= \max\Big\{v(s) - F^\star,\ \max_{i\in\mathcal{A}}\Big[-c_i + \sum_{z\in\mathcal{Y}} P_i(z \mid s)\big(J(s\vee z) - F^\star\big)\Big]\Big\} \\ &= \max\Big\{v(s) - F^\star,\ \max_{i\in\mathcal{A}}\Big[-c_i + \sum_{z\in\mathcal{Y}} P_i(z \mid s)\, B(s\vee z)\Big]\Big\}. \end{aligned} \tag{B-15}$$

We next show that (B-15) is exactly the agent Bellman equation (B-13).

Fix any state $s \in \mathcal{Y}$, and write

$$C(s) := \max_{i\in\mathcal{A}}\Big[-c_i + \sum_{z\in\mathcal{Y}} P_i(z \mid s)\, B(s\vee z)\Big].$$

Then (B-15) becomes

$$B(s) = \max\{v(s) - F^\star,\ C(s)\}.$$

Since $B(s) \geq 0$ by (B-14), the maximum on the right-hand side is nonnegative. We claim that

$$\max\{v(s) - F^\star,\ C(s)\} = \max\{(v(s) - F^\star)^+,\ C(s)\}. \tag{B-16}$$

Indeed, there are two cases.

Case 1: $v(s) - F^\star \geq 0$. Then $(v(s) - F^\star)^+ = v(s) - F^\star$, so (B-16) is immediate.

Case 2: $v(s) - F^\star < 0$. Then $(v(s) - F^\star)^+ = 0$. Since

$$B(s) = \max\{v(s) - F^\star,\ C(s)\} \geq 0,$$

and the first term $v(s) - F^\star$ is negative, it must be that

$$C(s) \geq 0.$$

Therefore

$$\max\{v(s) - F^\star,\ C(s)\} = C(s) = \max\{0, C(s)\} = \max\{(v(s) - F^\star)^+,\ C(s)\}.$$

This proves (B-16).

Substituting (B-16) into (B-15), we obtain

$$B(s) = \max\Big\{(v(s) - F^\star)^+,\ \max_{i\in\mathcal{A}}\Big[-c_i + \sum_{z\in\mathcal{Y}} P_i(z \mid s)\, B(s \vee z)\Big]\Big\}, \qquad s \in \mathcal{Y}.$$

Thus B is a bounded solution of the Bellman equation (B-13). By the uniqueness established in Step 1 based on Lemma B-1, we have

$$A^\star(s) = B(s) = J(s) - F^\star \qquad \forall s \in \mathcal{Y}.$$

Step 3: Coincidence of continuation argmax sets.

Fix any incumbent $s \in \mathcal{Y}$ and any action $i \in \mathcal{A}$. Using Step 2, the agent's continuation payoff from action i under $w^\star$ is

$$\begin{aligned} -c_i + \sum_{z\in\mathcal{Y}} P_i(z \mid s)\, A^\star(s \vee z) &= -c_i + \sum_{z\in\mathcal{Y}} P_i(z \mid s)\big(J(s \vee z) - F^\star\big) \\ &= \Big[-c_i + \sum_{z\in\mathcal{Y}} P_i(z \mid s)\, J(s \vee z)\Big] - F^\star. \end{aligned}$$

Subtracting the same constant $F^\star$ from all continuation payoffs does not change the set of maximizing actions. Hence, for every incumbent s, the set of continuation actions maximizing the agent's Bellman equation under $w^\star$ coincides exactly with the set of continuation actions maximizing the first-best Bellman equation (12).

Step 4: The induced equilibrium policy is first-best.

Step 3 identifies the agent's best continuation actions: conditional on continuing, the actions that maximize the agent's payoff under $w^\star$ are exactly the actions that maximize the first-best continuation value. This does not yet prove that the induced policy is first-best, because the policy must also specify whether the agent stops or continues. We check the stop-versus-continue margin directly.

Fix any incumbent $s \in \mathcal{Y}$.

Case 1: the first-best strictly stops at s. Then

$$v(s) > \max_{i \in \mathcal{A}} \Big[-c_i + \sum_{z \in \mathcal{Y}} P_i(z \mid s)\, J(s \vee z) \Big].$$

Hence, by the first-best Bellman equation,

$$J(s) = v(s).$$

Subtracting $F^\star$ from the strict inequality above gives

$$v(s) - F^\star > \max_{i \in \mathcal{A}} \Big[-c_i + \sum_{z \in \mathcal{Y}} P_i(z \mid s) \big(J(s \vee z) - F^\star\big) \Big].$$

Using Step 2, this becomes

$$v(s) - F^\star > \max_{i \in \mathcal{A}} \Big[-c_i + \sum_{z \in \mathcal{Y}} P_i(z \mid s)\, A^\star(s \vee z) \Big].$$

Moreover, since $J(s) = v(s) \geq J(0) = F^\star$, we have

$$w^\star(s) = (v(s) - F^\star)^+ = v(s) - F^\star = A^\star(s).$$

Therefore

$$w^\star(s) > \max_{i \in \mathcal{A}} \Big[-c_i + \sum_{z \in \mathcal{Y}} P_i(z \mid s)\, A^\star(s \vee z) \Big],$$

so the agent strictly prefers stopping. Thus the induced choice agrees with the first-best.

Case 2: the first-best strictly continues at s. Then

$$J(s) = \max_{i \in \mathcal{A}} \Big[-c_i + \sum_{z \in \mathcal{Y}} P_i(z \mid s)\, J(s \vee z) \Big] > v(s).$$

Choose any action $i^\star$ attaining the continuation maximum. Then

$$-c_{i^\star} + \sum_{z \in \mathcal{Y}} P_{i^\star}(z \mid s)\, J(s \vee z) = J(s).$$

Using Step 2, namely $A^\star(\cdot) = J(\cdot) - F^\star$, we obtain

$$-c_{i^\star} + \sum_{z \in \mathcal{Y}} P_{i^\star}(z \mid s)\, A^\star(s \vee z) = -c_{i^\star} + \sum_{z \in \mathcal{Y}} P_{i^\star}(z \mid s) \big(J(s \vee z) - F^\star\big)$$

$$= \Big[- c_{i^\star} + \sum_{z \in \mathcal{Y}} P_{i^\star}(z \mid s)\, J(s \vee z) \Big] - F^\star$$
$$= J(s) - F^\star$$
$$= A^\star(s).$$

Thus, action $i^\star$ gives the agent continuation payoff $A^\star(s)$. By the definition of the continuation payoff, action $i^\star$ is agent-optimal, taking into account the option to stop. Therefore, under principal-favor tie-breaking, the agent continues with action $i^\star$.

Case 3: the first-best is indifferent between stopping and continuing at s. Then

$$J(s) = v(s) = \max_{i \in \mathcal{A}} \Big[- c_i + \sum_{z \in \mathcal{Y}} P_i(z \mid s)\, J(s \vee z) \Big].$$

Because $J(s) = v(s) \geq F^\star$, Step 2 implies

$$A^\star(s) = J(s) - F^\star = v(s) - F^\star = w^\star(s).$$

Thus, stopping is agent-optimal. Since stopping is also first-best optimal in the present case, the induced choice is first-best.

Step 5: Equilibrium payoffs under $w^\star$.

At the initial incumbent $s_0 = 0$, Step 2 gives the agent's equilibrium payoff as

$$u_A(w^\star) = A^\star(0) = J(0) - F^\star = 0.$$

Since Step 4 shows that the induced policy is first-best, the total surplus generated from $s_0 = 0$ equals $J(0)$. Therefore, the principal's equilibrium payoff is

$$u_P(w^\star) = J(0).$$

Step 6: Optimality of $w^\star$.

Under any contract w and any best response, the agent can always stop immediately, so $u_A(w) \geq 0$. Moreover, the total surplus cannot exceed the first-best value: $u_P(w) + u_A(w) \leq J(0)$. Combining these two inequalities gives $u_P(w) \leq J(0)$. By Step 5, $u_P(w^\star) = J(0)$, so $w^\star$ attains this first-best upper bound and is therefore second-best optimal. □

Proof of Lemma B-1. Fix any history whose current incumbent is s. By definition of the incumbent, every realized raw outcome in the history is at most s. Since w is nondecreasing, for every realized outcome $y \leq s$ we have

$$w(y) \leq w(s).$$

Therefore, if the agent stops at that history, her optimal report is the incumbent s, and the stopping payoff is exactly $w(s)$. Since w is nondecreasing, all future payments are at most $w(m)$; since

sampling costs are nonnegative, no continuation policy can give the agent expected payoff above $w(m)$. Stopping immediately gives payoff $w(s) \geq 0$. Hence $0 \leq A_w(s) \leq w(m)$, so the continuation value is bounded. Moreover, $A_w(s)$ must satisfy (B-12). It remains to prove uniqueness among bounded solutions. We proceed by backward induction on the incumbent state.

At the top state $s = m$, for every raw draw $z \in \mathcal{Y}$ we have $m \vee z = m$. Hence, for every action $i \in \mathcal{A}$,

$$\sum_{z \in \mathcal{Y}} P_i(z \mid m)\, A_w(m \vee z) = \sum_{z \in \mathcal{Y}} P_i(z \mid m)\, A_w(m) = A_w(m).$$

Substituting into (B-12) gives

$$A_w(m) = \max\{w(m), \max_{i \in \mathcal{A}}[-c_i + A_w(m)]\}.$$

Since $c_i > 0$ for every i, we have $-c_i + A_w(m) < A_w(m)$, so necessarily

$$A_w(m) = w(m).$$

Thus the value at the top state is uniquely determined.

Now fix $s < m$, and suppose that $A_w(z)$ has already been uniquely determined for all $z > s$. Set

$$x := A_w(s), \qquad r_i(s) := \sum_{z \leq s} P_i(z \mid s), \qquad B_i(s) := \sum_{z > s} P_i(z \mid s)\, A_w(z).$$

Since the values $A_w(z)$ for $z > s$ are already known, each $B_i(s)$ is a known constant. Moreover,

$$\sum_{z \in \mathcal{Y}} P_i(z \mid s)\, A_w(s \vee z) = r_i(s)x + B_i(s).$$

Therefore x must satisfy

$$x = T_s(x) := \max\Big\{w(s),\ \max_{i \in \mathcal{A}}\big[-c_i + r_i(s)x + B_i(s)\big]\Big\}. \tag{B-17}$$

If $r_i(s) = 1$, then $P_i(z \mid s) = 0$ for all $z > s$, so $B_i(s) = 0$, and the corresponding branch in (B-17) is

$$-c_i + x < x.$$

Hence such a branch can never attain the maximum at a fixed point. Therefore the fixed points of T_s coincide with the fixed points of the reduced map

$$\widetilde{T}_s(x) := \max\Big\{w(s),\ \max_{i:\, r_i(s) < 1}\big[-c_i + r_i(s)x + B_i(s)\big]\Big\},$$

with the convention that if $\{i : r_i(s) < 1\} = \emptyset$, then $\widetilde{T}_s(x) \equiv w(s)$.

Now define

$$\rho_s := \begin{cases} \max_{i:\, r_i(s)<1} r_i(s), & \text{if } \{i : r_i(s) < 1\} \neq \emptyset, \\ 0, & \text{otherwise.} \end{cases}$$

Then $\rho_s < 1$. Since each branch of $\widetilde{T}_s$ is affine with slope at most ρ_s, and the maximum of finitely many ρ_s-Lipschitz functions is again ρ_s-Lipschitz, we have

$$|\widetilde{T}_s(x) - \widetilde{T}_s(x')| \le \rho_s |x - x'| \qquad \text{for all } x, x' \in \mathbb{R}.$$

Thus $\widetilde{T}_s$ is a contraction on $\mathbb{R}$. By Banach's fixed-point theorem, it has a unique fixed point. Hence T_s also has a unique fixed point, which uniquely determines $A_w(s)$.

Backward induction from $s = m$ down to $s = 0$ therefore shows that (B-12) has a unique bounded solution. □

Proof of Proposition 7. Let T be the Bellman operator, defined on bounded functions $f : \mathcal{Y} \to \mathbb{R}$, by

$$(Tf)(s) := \max\Big\{v(s),\ \max_{i\in\mathcal{A}}\Big[-c_i + \sum_{z\in\mathcal{Y}} P_i(z \mid s)\, f(s \vee z)\Big]\Big\}, \qquad s \in \mathcal{Y}. \tag{B-18}$$

Since v is strictly increasing, it is nondecreasing. Applying Lemma B-1 with $w = v$, the first-best Bellman equation has a unique bounded solution. Equivalently, J is the unique bounded fixed point of T, i.e.

$$J = TJ.$$

We first show that if f is bounded and nondecreasing, then Tf is also nondecreasing. Fix any such function f, fix an action $i \in \mathcal{A}$, and fix incumbents $s' \ge s$. Recall that

$$X_i(s) := s \vee Z_{i,s}, \qquad Z_{i,s} \sim P_i(\cdot \mid s).$$

Because $X_i(s)$ takes values in $\mathcal{Y} = \{0, 1, \dots, m\}$, we can write

$$f(X_i(s)) = f(0) + \sum_{y=1}^{X_i(s)} \big(f(y) - f(y-1)\big). \tag{B-19}$$

Taking expectations and rearranging the finite sum gives

$$\mathbb{E}\big[f(X_i(s))\big] = f(0) + \sum_{y=1}^{m} \big(f(y) - f(y-1)\big)\, \Pr\big(X_i(s) \ge y\big). \tag{B-20}$$

Since f is nondecreasing, each increment $f(y) - f(y-1)$ is nonnegative. Combining (B-20) with the FOSD assumption (FOSD), we obtain

$$\mathbb{E}\big[f(X_i(s'))\big] \ge \mathbb{E}\big[f(X_i(s))\big].$$

Equivalently,

$$\sum_{z\in\mathcal{Y}} P_i(z \mid s')\, f(s' \vee z) \;\geq\; \sum_{z\in\mathcal{Y}} P_i(z \mid s)\, f(s \vee z).$$

Therefore, for each action i, the continuation payoff

$$-c_i + \sum_{z\in\mathcal{Y}} P_i(z \mid s)\, f(s \vee z)$$

is nondecreasing in s. Returning to the definition of T in (B-18), the stopping term $v(s)$ is also nondecreasing. Hence the maximum over the stopping payoff and all action-continuation payoffs is nondecreasing, so

$$(Tf)(s') \geq (Tf)(s) \qquad \text{whenever } s' \geq s.$$

Thus Tf is nondecreasing.

Now define the value-iteration sequence

$$J^{(0)}(s) := v(s), \qquad J^{(n+1)} := TJ^{(n)} \quad \text{for } n \geq 0.$$

Because $J^{(0)} = v$ is nondecreasing and T preserves nondecreasing functions, induction gives that each $J^{(n)}$ is nondecreasing. Equivalently,

$$J^{(n)}(s') \geq J^{(n)}(s) \qquad \forall n, \quad s' \geq s. \tag{B-21}$$

Thus value iteration starts from a nondecreasing lower envelope and never leaves the class of nondecreasing functions. The remaining step is only to verify that the pointwise limit of these iterates is the fixed point J.

Next, T is monotone as an operator: if $f \leq g$ pointwise, then $Tf \leq Tg$ pointwise. Also, from the Bellman equation we have $v \leq J$ pointwise, because stopping is always feasible. Hence

$$J^{(n)} = T^n v \leq T^n J = J \qquad \forall n \geq 0,$$

where T^n denotes the n-fold iterate of the Bellman operator T. Moreover,

$$J^{(1)} = Tv \geq v = J^{(0)},$$

and by monotonicity of T this implies

$$J^{(n+1)} \geq J^{(n)} \qquad \forall n \geq 0.$$

Hence, for each fixed state $s \in \mathcal{Y}$, the real sequence

$$\{J^{(n)}(s)\}_{n\geq 0}$$

is nondecreasing and bounded above by $J(s)$. Therefore the limit

$$J^{\infty}(s) := \lim_{n\to\infty} J^{(n)}(s)$$

exists for every $s \in \mathcal{Y}$. Equivalently, the pointwise limit

$$J^{\infty} := \lim_{n\to\infty} J^{(n)}$$

is well defined, and it satisfies

$$J^{\infty}(s) \le J(s) \qquad \forall s \in \mathcal{Y}.$$

Because $\mathcal{Y}$ and $\mathcal{A}$ are finite, T is continuous under the pointwise increasing limit $J^{(n)} \uparrow J^{\infty}$. We spell out the argument pointwise. Fix $s \in \mathcal{Y}$. Then

$$\begin{aligned}
(TJ^{\infty})(s) &= \max\left\{v(s),\ \max_{i\in\mathcal{A}}\left[-c_i + \sum_{z\in\mathcal{Y}} P_i(z \mid s)\, J^{\infty}(s\vee z)\right]\right\} \\
&= \max\left\{v(s),\ \max_{i\in\mathcal{A}}\left[-c_i + \sum_{z\in\mathcal{Y}} P_i(z \mid s)\, \lim_{n\to\infty} J^{(n)}(s\vee z)\right]\right\} \\
&= \max\left\{v(s),\ \max_{i\in\mathcal{A}}\left[-c_i + \lim_{n\to\infty}\sum_{z\in\mathcal{Y}} P_i(z \mid s)\, J^{(n)}(s\vee z)\right]\right\} \\
&= \lim_{n\to\infty} \max\left\{v(s),\ \max_{i\in\mathcal{A}}\left[-c_i + \sum_{z\in\mathcal{Y}} P_i(z \mid s)\, J^{(n)}(s\vee z)\right]\right\} \\
&= \lim_{n\to\infty} (TJ^{(n)})(s) \\
&= \lim_{n\to\infty} J^{(n+1)}(s) \\
&= J^{\infty}(s).
\end{aligned}$$

The first equality is the definition of T in (B-18). The second uses $J^{\infty} = \lim_n J^{(n)}$. The third uses finiteness of $\mathcal{Y}$ to pass the limit through the finite sum. The fourth uses finiteness of $\mathcal{A}$ to pass the limit through the finite maximum over stopping and actions. The fifth returns to the definition of T, the sixth uses the value-iteration recursion $J^{(n+1)} = TJ^{(n)}$, and the last uses the definition of J^{∞}. Hence $TJ^{\infty} = J^{\infty}$, so J^{∞} is a bounded fixed point of T. By uniqueness of the bounded fixed point, $J^{\infty} = J$. Now apply (B-21) for $n \to \infty$, we have $J(s') \ge J(s)$ for all $s' \ge s$. This concludes the proof. □

Proof of Example 1. Fix $s' \ge s$. Since

$$X_i(s) = \min\{m,\ s + D_i^{+}\},$$

and the right-hand side is nondecreasing in s for every realization of D_i, we have

$$X_i(s') \ge X_i(s) \qquad \text{almost surely.}$$

Therefore, for every threshold $y \in \mathcal{Y}$,

$$\Pr\left(X_i(s') \geq y\right) \geq \Pr\left(X_i(s) \geq y\right),$$

which is exactly the FOSD condition (FOSD). The same argument applies action by action. Hence, if each action is generated by such a local-refinement shock D_i, the assumption in Proposition 7 holds. □

B.3.3 Proof of Section 6.3

Proof of Theorem 3. We proceed in four steps.

Step 1: A global upper bound on the multi-agent second best.

Obviously,

$$\mathsf{SB}_P(\mathbf{K}) \leq \mathsf{FB}_P(\mathbf{K}).$$

Relaxing the caps can only enlarge the first-best feasible set, hence

$$\mathsf{FB}_P(\mathbf{K}) \leq \mathsf{FB}_P(\boldsymbol{\infty}),$$

where $\boldsymbol{\infty}$ denotes the cap vector with all entries infinite.

In the benchmark $\mathsf{FB}_P(\boldsymbol{\infty})$, the principal directly controls search over the pooled action set

$$\widetilde{\mathcal{A}} := \{(g,i) : g \in \mathcal{G},\ i \in \mathcal{A}_g\},$$

where pooled action (g,i) has cost c_{gi} and outcome distribution p_{gi}. Since the principal ultimately adopts only one outcome, this is exactly the single-agent self-search problem with action set $\widetilde{\mathcal{A}}$, contract $w = v$, and infinite horizon. By Proposition 2(ii)(b),

$$\mathsf{FB}_P(\boldsymbol{\infty}) = \max_{(g,i)\in\widetilde{\mathcal{A}}} \zeta_{gi} = \max_{g\in\mathcal{G}} \bar{\zeta}_g = \bar{\zeta}_1.$$

Therefore,

$$\mathsf{SB}_P(\mathbf{K}) \leq \mathsf{FB}_P(\mathbf{K}) \leq \mathsf{FB}_P(\boldsymbol{\infty}) = \bar{\zeta}_1. \tag{B-22}$$

Step 2: Threshold-reaching events under the sequential debt contract.

For each agent $g \in \mathcal{G}$, fix an independent sequence of draws

$$(Y_{gk})_{k\geq 1} \qquad \text{with} \qquad Y_{gk} \sim p_{g,\bar{i}_g}.$$

Define the event

$$A_g := \left\{\exists\, k \leq K_g \text{ such that } v(Y_{gk}) \geq \bar{\zeta}_g\right\}.$$

By Lemma 2, under the debt contract with face value $\bar{\zeta}_g$, agent g's best response is to keep sampling the single action $\bar{i}_g$ until either she observes a draw with value at least $\bar{\zeta}_g$, or her cap is exhausted.

Hence, if agent g is reached under the sequential debt contract, then:

(a) on A_g, she stops at the first draw whose value is at least $\bar{\zeta}_g$, and this reported outcome satisfies

$$v(\hat{Y}_g) - w_g^{\textsf{seq-debt}}(\hat{Y}_g) = \bar{\zeta}_g;$$

(b) on A_g^c, she never reaches her debt threshold and thus every realized wage is zero.

Moreover, the events $(A_g)_{g\in\mathcal{G}}$ are independent across agents, and

$$\Pr(A_g) = 1 - (1 - q_g)^{K_g}, \tag{B-23}$$

with the convention $(1 - q_g)^{\infty} := 0$.

Step 3: A lower bound on the payoff of the sequential debt contract.

For each $g \in \mathcal{G}$, define the disjoint event

$$B_g := A_g \cap \bigcap_{\ell<g} A_\ell^c.$$

We claim that on B_g, the principal's realized payoff under the sequential debt contract is at least $\bar{\zeta}_g$.

Fix a sample path in B_g, and consider the execution of the sequential debt contract on that path. Under the fixed order $1, 2, \ldots, G$, after t sponsored agents, we have $\mathcal{S}_t = \{1, \ldots, t\}$ and

$$X_t = \max_{r\in\mathcal{S}_t} v(\hat{Y}_r).$$

If the principal stops before sponsoring agent g, then there exists some $\ell < g$ at which she stops. Since the sample path lies in A_r^c for every $r \le \ell$, no sponsored agent has reached her own debt threshold, so all wages paid up to round ℓ are zero. Thus the stopping rule implies

$$X_\ell \ge \bar{\zeta}_{\ell+1}.$$

Since $\ell < g$ and $\bar{\zeta}_1 \ge \bar{\zeta}_2 \ge \cdots \ge \bar{\zeta}_G$,

$$\bar{\zeta}_{\ell+1} \ge \bar{\zeta}_g.$$

By (13), the principal's realized payoff is X_ℓ, and hence is at least $\bar{\zeta}_g$.

If instead the principal does not stop before round g, then agent g is reached. Because the sample path lies in A_g, agent g reaches her threshold $\bar{\zeta}_g$ within her cap. Her debt contract then gives the principal net payoff

$$v(\hat{Y}_g) - w_g^{\textsf{seq-debt}}(\hat{Y}_g) = \bar{\zeta}_g.$$

All earlier wages are zero. Moreover, since the principal did not stop before round g, $X_{g-1} < \bar{\zeta}_g$ when $g > 1$, so no earlier report gives the principal more than agent g's threshold report. Finally, $\bar{\zeta}_g \ge \bar{\zeta}_{g+1}$, so the outer stopping rule stops after round g. Hence the principal adopts agent g's

report and obtains realized payoff exactly $\bar{\zeta}_g$.

Thus, on every sample path in B_g, the principal's realized payoff is at least $\bar{\zeta}_g$. Since the events $(B_g)_{g=1}^G$ are disjoint, we obtain

$$U_P^{\text{seq-debt}}(\mathbf{K}) \geq \sum_{g=1}^{G} \bar{\zeta}_g \Pr(B_g). \tag{B-24}$$

By independence and (B-23),

$$\Pr(B_g) = \Pr(A_g) \prod_{\ell<g} \Pr(A_\ell^c) = \Big(1-(1-q_g)^{K_g}\Big) \prod_{\ell<g} (1-q_\ell)^{K_\ell}.$$

Substituting into (B-24), we get

$$U_P^{\text{seq-debt}}(\mathbf{K}) \geq \sum_{g=1}^{G} \bar{\zeta}_g \Big(1-(1-q_g)^{K_g}\Big) \prod_{\ell<g} (1-q_\ell)^{K_\ell}. \tag{B-25}$$

Combining (B-25) with (B-22) yields

$$\frac{U_P^{\text{seq-debt}}(\mathbf{K})}{\mathsf{SB}_P(\mathbf{K})} \geq \frac{1}{\bar{\zeta}_1} \sum_{g=1}^{G} \bar{\zeta}_g \Big(1-(1-q_g)^{K_g}\Big) \prod_{\ell<g} (1-q_\ell)^{K_\ell},$$

which proves (17).

To prove (18), let $m^\star := \max \mathcal{G}^*$. Since the agents are ordered by weakly decreasing $\bar{\zeta}_g$, we have

$$\mathcal{G}^* = \{1, \dots, m^\star\}, \qquad \bar{\zeta}_g = \bar{\zeta}_1 \text{ for all } g \leq m^\star.$$

Hence

$$\begin{aligned}
\frac{1}{\bar{\zeta}_1} \sum_{g=1}^{G} \bar{\zeta}_g \Big(1-(1-q_g)^{K_g}\Big) \prod_{\ell<g} (1-q_\ell)^{K_\ell} &\geq \sum_{g=1}^{m^\star} \Big(1-(1-q_g)^{K_g}\Big) \prod_{\ell<g} (1-q_\ell)^{K_\ell} \\
&= \sum_{g=1}^{m^\star} \Pr(B_g) \\
&= \Pr\left(\bigcup_{g=1}^{m^\star} B_g\right) \\
&= 1 - \prod_{g=1}^{m^\star} (1-q_g)^{K_g} \\
&= 1 - \prod_{g\in\mathcal{G}^*} (1-q_g)^{K_g},
\end{aligned}$$

which proves (18).

Step 4: The infinite-cap cases.

Suppose

$$g_0 := \min\{g \in \mathcal{G} : K_g = \infty\}$$

exists. Since $q_{g_0} > 0$, the event A_{g_0} occurs almost surely when $K_{g_0} = \infty$. We claim that, pathwise, the principal's realized payoff under the sequential debt contract is at least $\bar{\zeta}_{g_0}$.

Fix any realized sample path. If the principal stops before sponsoring agent g_0, then there exists some $\ell < g_0$ at which she stops. Let $X_\ell = \max_{r \le \ell} v(\hat{Y}_r)$. If agent ℓ reaches her own debt threshold, then no earlier stop has occurred, so all earlier wages are zero and the principal's realized payoff is $\bar{\zeta}_\ell \ge \bar{\zeta}_{g_0}$. If agent ℓ does not reach her own debt threshold, then, as above, all wages paid up to round ℓ are zero and the stopping rule implies

$$X_\ell \ge \bar{\zeta}_{\ell+1} \ge \bar{\zeta}_{g_0},$$

and again the realized payoff is at least $\bar{\zeta}_{g_0}$.

If the principal does not stop before round g_0, then agent g_0 is reached. Because A_{g_0} occurs almost surely, agent g_0 eventually reaches her debt threshold. At that point her debt contract gives the principal net payoff $\bar{\zeta}_{g_0}$, all earlier reports are below the stopping cutoff $\bar{\zeta}_{g_0}$, and $\bar{\zeta}_{g_0} \ge \bar{\zeta}_{g_0+1}$ (with $\bar{\zeta}_{G+1} = 0$ if $g_0 = G$). Thus the outer policy stops and the realized payoff is exactly $\bar{\zeta}_{g_0}$.

Thus, on every sample path, the principal's realized payoff is at least $\bar{\zeta}_{g_0}$. Therefore,

$$U_P^{\mathsf{seq\text{-}debt}}(\mathbf{K}) \ge \bar{\zeta}_{g_0}.$$

Combining this with (B-22) yields

$$\frac{U_P^{\mathsf{seq\text{-}debt}}(\mathbf{K})}{\mathsf{SB}_P(\mathbf{K})} \ge \frac{\bar{\zeta}_{g_0}}{\bar{\zeta}_1},$$

which proves (19).

Finally, if $g_0 \in \mathcal{G}^*$, then $\bar{\zeta}_{g_0} = \bar{\zeta}_1$, so

$$U_P^{\mathsf{seq\text{-}debt}}(\mathbf{K}) \ge \bar{\zeta}_1.$$

Since always

$$U_P^{\mathsf{seq\text{-}debt}}(\mathbf{K}) \le \mathsf{SB}_P(\mathbf{K}) \le \bar{\zeta}_1,$$

we conclude that

$$U_P^{\mathsf{seq\text{-}debt}}(\mathbf{K}) = \mathsf{SB}_P(\mathbf{K}) = \bar{\zeta}_1.$$

This completes the proof. □